A DISSERTATION



THE ZOMBIE WHO BEGOT HIMSELF

Autogenesis, Aseity, and the Abaliety of Persons in Robert A. Heinlein's "—All You Zombies—"

OCTOBER 2026



Vergine Madre, figlia del tuo figlio,
umile e alta più che creatura,
termine fisso d'etterno consiglio.

DANTE *Paradiso* XXXIII.1–3

•

Quod cupio mecum est: inopem me copia fecit.

OVID *Metamorphoses* III.466

•

We know no time when we were not as now;
Know none before us, self-begot, self-rais'd
By our own quick'ning power.

MILTON *Paradise Lost* V.859–861

•

Non tamen invenitur, nec est possibile, quod aliquid sit causa efficiens sui ipsius; quia sic esset prius seipso, quod est impossibile.

AQUINAS *Summa Theologiae* I, q. 2, a. 3

ABSTRACT

This dissertation advances a new interpretation of Robert A. Heinlein's "—All You Zombies—" (composed 1958; published 1959), a text conventionally received either as the purest specimen of the self-consistent causal loop in speculative fiction or as a biographical symptom of its author's sexual and political preoccupations. Against both receptions, it argues that the story constitutes a rigorous narrative experiment in the predicate of *aseity*—self-derivation—transposed from its theological home to a finite, embodied, sexed human subject by means of a closed timelike genealogy, and that the notorious closing apostrophe, in which the narrator relegates the reader to the class of "zombies," is the hermeneutic key that converts a logical puzzle into a theory of the person.

Part I reconstructs the genealogies of the story's governing ideas: the geometrization of time from Hinton, Wells, and Minkowski through Gödel's rotating cosmology, Wheeler's one-electron universe, and the Novikov self-consistency principle; Heinlein's documented debts to Korzybski's General Semantics, to Twain's late determinism and solipsism, and to Cabell's ironic metaphysics of recurrence; the scholastic and early-modern theology of aseity and *causa sui* (Anselm, Aquinas, Descartes, Spinoza) together with its heterodox shadows in Sethian Gnosticism, Marian paradox, modalist Trinitarianism, and Milton's Satan; and the iconographic lineage of the ouroboros, the androgyne, Narcissus, and Tiresias, culminating in Eliot's "foresuffering" witness.

Part II subjects the text to formal, biopolitical, psychoanalytic, metaphysical, and comparative analysis. It demonstrates, inter alia, that the timestamped *syuzhet* enacts a McTaggartian B-series governed by Lewisian personal time; that the embedded confession reproduces the Augustinian scene of narration before an eternal auditor; that the narrator's body is a non-compossible substrate whose genomic fixed point would require a Wrightian coefficient of inbreeding of unity and the total absence of de novo mutation; that the abduction plot literalizes the postwar seizure of unwed mothers' infants; that the story's psychic architecture is an inverted homologue of Schreber's *Memoirs*; and that its genealogical relation is formally a Quine atom in non-well-founded set theory.

Part III synthesizes these findings into the **Abaliety Thesis**: personhood is constitutively *ab alio*—derived from another—such that a subject who attains

genealogical closure (here termed the **Heinlein Limit**) forfeits recognition, natality, and alterity-bearing time, and thereby occupies precisely the position its apostrophe projects upon others (the **Apostrophic Inversion**). The story is shown to be a *reductio* of what the dissertation names the **Fallacy of Autogenetic Division**: the transfer of self-origination from collectivities—species, civilizations, traditions—where it may be legitimately predicated, to individuals, where it issues in solipsistic catastrophe. The fallacy is traced through possessive individualism, the American mythology of self-making, and contemporary recursive information systems. The dissertation concludes that the answer to the narrator's final question is the definition of the person: we came from one another.

NOTE ON METHOD AND CITATION

The method is constellational in Benjamin's sense—disparate materials are arranged until a figure legible in none of them singly becomes visible—disciplined by three formal instruments: the metaphysics of time (McTaggart, Lewis, Sider), the mathematics of self-reference (Kleene, Aczel, Barwise and Moss), and population genetics (Wright, Muller). Classical, patristic, and scholastic sources are cited by standard divisions; modern sources by author and date, with full references in the Selected Bibliography. The primary text, which survives in numerous editions and anthologies, is cited by narrative segment rather than page, and its events are paraphrased rather than excerpted. Where the dissertation identifies synchronies among contemporaneous texts, it claims constellation rather than influence unless influence is documented.

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THE PROBLEM OF THE SELF-CAUSED SUBJECT

§ I

The Text and Its Occasion

"—All You Zombies—" was, by the author's own records as reconstructed in William H. Patterson's authorized biography (2014), composed in a single day in July 1958, declined (by most accounts) by *Playboy*, and published in the March 1959 number of *The Magazine of Fantasy and Science Fiction*. Its first book appearance, in the Gnome Press collection *The Unpleasant Profession of Jonathan Hoag* (1959), placed it beside "They" (1941), a confirmed solipsist's fable; "—And He Built a Crooked House—" (1941), a Hintonian tesseract made habitable; and the title novella (1942), in which the perceived world is a work of art under critical revision. The collection is thus, whatever its commercial genesis, a metaphysical quartet on solipsism, dimensionality, and the ontological standing of the perceived world, and the 1958 story is its terminal statement. The story was adapted by Michael and Peter Spierig as *Predestination* (2014), whose added antagonist thickens the loop without altering its logic.

The months of composition are dense with what this dissertation will treat as a constellation rather than a chain of influence. Hugh Everett's relative-state formulation of quantum mechanics had appeared in *Reviews of Modern Physics* in July 1957. Hannah Arendt's *The Human Condition*, with its doctrine of natality and plurality, and Hans Jonas's *The Gnostic Religion*, with its existentialist reading of the self-begotten demiurge, were both published in 1958. Jacques Lacan drafted his écrit on the foreclosure of the Name-of-the-Father in psychosis across the winter of 1957–58. Fritz Leiber's *The Big Time*, a Change War novel staffed by recuperative "Entertainers," took the Hugo in 1958; Alfred Bester's "The Men Who Murdered Mohammed," in which each time traveler is sealed inside a private past, appeared in the same magazine in October 1958. Reichenbach's *The Philosophy of Space and Time*, with its discussion of closed causal chains and the self-encountering traveler, appeared in English translation that year. In the same season Heinlein himself was drafting *Starship Troopers*, a novel of military recruitment as the forge of identity, and continuing the long gestation of *Stranger in a Strange Land*, whose apostrophe "Thou art God" would invert the valence of his "you zombies" within two years. The story stands at an intersection of physics, theology, psychoanalysis, and political anthropology that it registers with unusual compression.

§ II

The Fabula Reconstructed

The Russian Formalist distinction between *fabula* (the events in their causal-chronological order) and *syuzhet* (their order of presentation) is here not merely useful but constitutive, because the story's *fabula* is not a line. The *syuzhet* is a sequence of timestamped segments narrated by a bartender who is in fact an agent of a Temporal Bureau. In a New York bar in November 1970, a twenty-five-year-old writer of confession stories, published under the byline "The Unmarried Mother," wagers that his life story is stranger than any the bartender has heard. He recounts it. He was a foundling, deposited at a Cleveland orphanage in 1945 and raised as a girl, Jane. Plain, solitary, and ambitious, she sought admission to W.E.N.C.H.E.S., a quasi-military corps supplying sexual and social companionship to spacemen. In April 1963, aged seventeen by the story's own dates, she was seduced, impregnated, and abandoned by an older stranger; pregnancy disqualified her. At delivery by Caesarean section in 1964, surgeons discovered two complete sets of reproductive organs, removed the damaged female set, and reconstituted the patient as a man. The infant daughter was abducted from the nursery by an unknown man. The Unmarried Mother, now male, writes confessions and nurses a desire for revenge upon the seducer.

The bartender offers him the seducer. Using the Bureau's field equipment he conveys the young man to Cleveland in 1963, where the young man meets and seduces the girl Jane—his former self. The bartender proceeds to 1964, removes the infant from the hospital, carries her to 1945, and leaves her at the orphanage. He collects the young man in 1963, conveys him to the Bureau in the mid-1980s, and enlists him. He returns to the bar in 1970, minutes after his departure, and finally retires to the Bureau's subterranean base in 1993, where, examining the Caesarean scar on his own abdomen and the ouroboros ring on his own finger, he acknowledges that he is all of them: foundling, Jane, the Unmarried Mother, the seducer, the abductor, the recruiter. He closes by turning to the reader with the question that gives the story its title, an inquiry into the origins of everyone else, whom he designates as zombies, together with the assertion that in truth there is no one present but himself, alone in the dark.

In external time the decisive events are ordered 1945 (deposition), 1963 (seduction), 1964 (birth, surgery, abduction), 1970 (bar), mid-1980s (recruitment), 1993 (coda). In the person's *personal time*—the time of his own aging, memory, and causal agency, in David Lewis's sense—the order is: birth (1964)

→ childhood (1945–63) → seduction as Jane (1963) → delivery and surgery (1964) → life as the Unmarried Mother → the bar as patron (1970) → seduction as seducer (1963) → enlistment (1980s) → career → the bar as bartender (1970) → abduction (1964) → deposition (1945) → collection (1963) → recruitment (1980s) → coda (1993). The worldline is therefore not a circle but what this dissertation will call a **σ-topology**: a closed loop (from the infant's birth to the mother's delivery of that same infant, and, by a second and orientation-reversing circuit, to the father's act of conception) with a tail extending into the narrator's unclosed future. The loop is closed at the origin and open at the end; the subject is causally sealed at birth and existentially exposed at death. Much of what follows depends on this asymmetry.

§ III

The State of the Question

The story's reception has been bifurcated. In analytic philosophy it is the canonical exhibit of the consistent causal loop, the case that David Lewis's "The Paradoxes of Time Travel" (1976) renders intelligible through the distinction of personal from external time, and that Paul Horwich (1975, 1987), Paul Nahin (1993), and Nicholas J. J. Smith (1997) treat as the paradigm of improbable but non-contradictory coincidence. In literary criticism it has been read as craft (Alexei Panshin's *Heinlein in Dimension*, 1968), as an instance of the secular Calvinism George Slusser discerned in the oeuvre (1976, 1977), as a document of Cold War America (H. Bruce Franklin, 1980), and, more recently, within Farah Mendlesohn's comprehensive reassessment of Heinlein's sexual politics (2019). Constance Penley's "Time Travel, Primal Scene, and the Critical Dystopia" (1986) and David Wittenberg's *Time Travel: The Popular Philosophy of Narrative* (2013) supply the psychoanalytic and narratological frames on which Chapters 5 and 7 build. Recent intersex and trans readings have reclaimed Jane's embodiment from the caricature that earlier criticism often made of it.

What no account has done is to treat the closing apostrophe as structurally decisive, or to integrate the theological doctrine of aseity, the physics of closed timelike curves, the psychoanalysis of psychotic foreclosure, and the political anthropology of the self-made individual into a single theory of the person that the story both presupposes and refutes. Philosophical readings stop at consistency; literary readings stop at symptom. The lacuna lies between.

§ IV

Thesis

The central claim is the **Abaliety Thesis**. Scholastic philosophy distinguished *aseitas*, existence from oneself, attributed to God alone, from *abalietas*, existence from another, the condition of every creature. Heinlein's story effects a technological transfer of aseity to a finite subject. It thereby isolates abaliety by subtraction, and demonstrates that abaliety is not a creaturely deficiency but the constitutive condition of personhood: to be a person is to come from another, to be addressed by another, and to be able to begin. Four subsidiary theses follow.

1. **The Heinlein Limit.** There is a limit case of self-origination at which a subject's entire efficient-causal ancestry converges upon its own temporal parts. At this limit, recognition, natality, and alterity-bearing time vanish together.
2. **The Apostrophic Inversion.** The closing epithet is semantically self-referential. On every relevant sense of the term—the Haitian *zonbi* as a body executing another's will, the Cold War conformist, the philosophical being without an inner relation to others—the narrator, not the reader, is the zombie.
3. **The Fallacy of Autogenetic Division.** Self-origination is legitimately predicated of collectivities (Marx's self-producing species, Kant's self-generating natural kind, Korzybski's time-binding civilization, Arendt's plurality) only on condition that it be denied of their members. The story is a *reductio* of the individualist error that transfers the collective predicate to the individual.
4. **Chronogenealogy.** The analysis of kinship in spacetimes admitting closed timelike curves is a legitimate formal discipline, with consequences for the logic of descent (non-well-foundedness), for population genetics (fixed-point constraints), and for the theory of narrative (the autonarratee).

§ V

Architecture

Part I (Chapters 1–4) reconstructs the genealogies of the story's ideas: physical, Heinleinian, theological, and iconographic. Part II (Chapters 5–11) offers formal, biopolitical, psychoanalytic, metaphysical, ethical, epistemological, and comparative analyses. Part III (Chapter 12) states the Abaliety Thesis formally, defends it against objections, and extends it to political theory and to recursive information systems. The Conclusion answers the narrator's question.



CHRONOMETRICS

The Geometrization of Time and the Physics of Closed Genealogy

The story presupposes a block universe in which the past is not altered but completed. This chapter reconstructs the lineage of that presupposition and shows that the story anticipates, with unusual exactness, the self-consistency constraints that relativistic physics would formalize three decades later.

§ 1.1

From Hinton's Tesseract to Minkowski's Block

Charles Howard Hinton's "What Is the Fourth Dimension?" (1880; pamphlet 1884) and his *Scientific Romances* (1884–86) proposed that time might be a fourth spatial dimension through which consciousness moves, and devised colored cubes for training a four-dimensional intuition. Wells's Time Traveller (1895) made the proposition narrative, treating duration as a dimension differing from the spatial three only in that consciousness travels along it. Hermann Minkowski's Cologne lecture "Raum und Zeit" (1908) gave the idea its physical form: space and time as aspects of a single four-dimensional manifold, the history of every body a *worldline*, causal influence confined within light cones. The resulting eternalism—the doctrine that past, present, and future events are equally real—is the metaphysical ground on which Heinlein builds. His "—And He Built a Crooked House—" (1941) had already rendered Hinton's tesseract as architecture. "—All You Zombies—" relocates the fourth dimension from the house to the person: the worldline itself becomes the dwelling, and it is folded.

§ 1.2

McTaggart and the Timestamp

J. M. E. McTaggart's "The Unreality of Time" (*Mind*, 1908) distinguished the A-series (past, present, future) from the B-series (earlier than, later than) and argued that time requires the former, which is contradictory. Whatever one makes of the argument, the distinction illuminates the story's form. Each segment is headed by a coordinate of the type "time, zone, date, place": a B-series designation, tenseless and public. The narration, however, unfolds in the narrator's own A-series, his lived present moving through his remembered past and anticipated future. The reader thus experiences a double inscription, reading a personal A-series beneath a heading in external B-series coordinates, and the segments are ordered by the former while labeled by the latter.

David Lewis's "The Paradoxes of Time Travel" (1976) supplies the instrument for resolving the apparent disorder. External time is time itself; personal time is the ordering of a traveler's stages by their causal and functional continuity (aging, memory, digestion, the accumulation of scar tissue). In ordinary lives the two coincide; in the traveler's they diverge. The story is an exercise in the divergence. McTaggart also posited a C-series, an ordering of

events without direction. The closed portion of the narrator's worldline realizes something McTaggart did not contemplate: a C-series with cyclic topology, an ordering without direction that is also without terminus at its origin.

§ 1.3

Interwar Popular Metaphysics: Dunne, Ouspensky, Priestley

The generation of writers who formed Heinlein's idiom absorbed time-metaphysics less from physicists than from popularizers. J. W. Dunne's *An Experiment with Time* (1927) and *The Serial Universe* (1934) argued from precognitive dreams to a regress of observers, each observing the time-dimension of the one below. P. D. Ouspensky's *Tertium Organum* (1912; English 1920), *A New Model of the Universe* (1931), and *Strange Life of Ivan Osokin* (1915) disseminated a doctrine of eternal recurrence in which a life, once lived, is relived identically unless consciousness intervenes. J. B. Priestley's "time plays"—*Dangerous Corner* (1932), *Time and the Conways* (1937), *I Have Been Here Before* (1937)—carried Dunne and Ouspensky to the West End. Heinlein's "Elsewhen" (1941), in which a professor's students travel among alternative timelines by mental discipline, belongs to this ambient. "—All You Zombies—" fuses the two popular schemata: Dunne's regress of observers (the bartender observing the young man observing his former self) and Ouspensky's recurrence (a life whose every term is fixed), stripped of Ouspensky's soteriological escape clause.

§ 1.4

Gödel's Universe and the Idealist Inference

Kurt Gödel's "An Example of a New Type of Cosmological Solutions of Einstein's Field Equations of Gravitation" (*Reviews of Modern Physics*, 1949) exhibited a rotating universe in which closed timelike curves pass through every point, following W. J. van Stockum's rotating dust cylinder (1937). In his contribution to Schilpp's *Albert Einstein: Philosopher-Scientist* (1949), Gödel drew the philosophical moral: if one may revisit one's past, the objective lapse of time is illusory, and Kant's doctrine of the ideality of time is vindicated. Einstein, replying in the same volume, conceded that the problem had troubled him from the construction of general relativity onward. Palle Yourgrau (1999, 2005) has reconstructed the argument and its neglect.

The story is a Gödelian thought experiment at the scale of a single life. Its narrator draws an idealist inference structurally identical to Gödel's, but dis-

placed in object. Gödel inferred from closed timelike curves the ideality of time; Heinlein's narrator infers, from his own closed genealogy, the ideality of *others*. This displacement—from the unreality of succession to the unreality of persons—is the first appearance of the Apostrophic Inversion, and it will recur in every chapter.

§ 1.5

Backward Causation in the Analytic 1950s

The decade of the story was also the decade in which analytic philosophy reopened the question of retrocausation. Michael Dummett's "Can an Effect Precede Its Cause?" and Antony Flew's reply of the same title (*Proceedings of the Aristotelian Society*, Supplementary Volume, 1954), followed by Max Black's "Why Cannot an Effect Precede Its Cause?" (*Analysis*, 1956), established the terms. Flew and Black urged the *bilking argument*: if B is alleged to cause an earlier A, one may observe whether A has occurred and then prevent B, refuting the causal claim. Dummett's later "Bringing About the Past" (1964), with its chief who dances after the hunt to secure his young men's courage, specified the epistemic conditions under which bilking is unavailable. Hans Reichenbach's *Philosophie der Raum-Zeit-Lehre* (1928) had already discussed closed causal chains and the traveler who meets his earlier self.

Heinlein's loop is constructed to be unbilkable. Every agent capable of bilking it possesses complete knowledge of its outcome, and that knowledge is itself memory of its having been completed. The seducer cannot refrain, because he remembers having been seduced; the abductor cannot refrain, because the infant must reach 1945 for the abductor to exist. Bilking is not prohibited by an external law but rendered motivationally unavailable by the identity of the potential bilker with the loop's product.

§ 1.6

One Electron, One Person: Wheeler, Feynman, Stueckelberg

In a telephone call of 1940, recounted in Richard Feynman's Nobel lecture (1965), John Archibald Wheeler proposed that all electrons have the same mass and charge because they are a single electron, weaving back and forth through time, its backward-running segments appearing as positrons. E. C. G. Stueckelberg (1941) and Feynman ("The Theory of Positrons," 1949) retained the interpretation of the positron as a backward-propagating electron while abandoning the cosmology, since it predicts equal numbers of electrons and positrons. The Wheeler–Feynman absorber theory (1945, 1949) had already admitted advanced as well as retarded solutions into classical electrodynamics.

The analogy to the story is exact in structure and instructive in its failure. Wheeler's single electron explains the identity of all electrons; the narrator's single self explains only himself. A one-electron universe leaves no surplus. A one-person universe leaves an enormous one: the six billion others whom the narrator's genealogy does not generate and therefore cannot explain. Call this the **Wheelerian surplus**. The final question of the story, an inquiry into where everyone else came from, is precisely the question that a one-particle cosmology answers and a one-person cosmology cannot.

§ 1.7

Self-Consistency and Its Discontents: Novikov, Thorne, Hawking, Deutsch

The physics of closed timelike curves acquired its modern form in the late 1980s. Michael Morris, Kip Thorne, and Ulvi Yurtsever (*Physical Review Letters*, 1988) showed that traversable wormholes, if they exist, can be converted into time machines. Igor Novikov proposed, and John Friedman and collaborators formalized (*Physical Review D*, 1990), the principle of self-consistency: the only solutions to the laws of physics that occur locally are those that are globally self-consistent. Fernando Echeverria, Gunnar Klinkhammer, and Thorne (1991) studied billiard balls entering a wormhole on trajectories that would knock their earlier selves off course (Polchinski's paradox) and found that consistent solutions always exist but are frequently non-unique: a single set of initial data admits several self-consistent histories. Stephen Hawking's chronology protection conjecture (1992) proposed that vacuum fluctuations destroy any time machine before it can form.

The story is a Novikov solution avant la lettre, and it exhibits the non-uniqueness problem in human form. Nothing in the initial data of 1945 determines that a Jane should exist; the history in which she exists is one consistent solution among others, including the trivial one in which she does not. The question "why this history?" has no answer internal to physics.

David Deutsch's quantum treatment (*Physical Review D*, 1991; with Michael Lockwood in *Scientific American*, 1994) replaced classical consistency with a fixed-point condition on density matrices, guaranteed to have a solution but possibly a mixed one, interpretable in Everettian terms as a braid of branches. Deutsch also identified the *knowledge paradox*: a causal loop can carry information—his example was an unproven theorem handed backward to its own author—that was never created by any process, violating the principle that knowledge arises only through evolutionary or creative processes. Jane's genome is such information. Chapter 6 develops the point quantitatively; Chapter 12 develops the Deutschian alternative as an Everettian therapy for autogenetic solipsism.

§ 1.8

Everett 1957: The Road Not Taken

Everett's "'Relative State' Formulation of Quantum Mechanics" (*Reviews of Modern Physics*, 1957) preceded the story by a year, though its popularization as the "many-worlds" interpretation awaited Bryce DeWitt (1970). The juxtaposition clarifies Heinlein's choice. The story elects a single-history, self-consistent topology, in which the only way to return to one's origin is to have always been there, and that election entails solitude. The Everettian alternative would pluralize the self into branch-relative persons, each deriving from another. Heinlein's late multiverse fiction, from *The Number of the Beast* (1980) onward, can be read as his belated passage from Novikovian closure to Everettian plurality.



THE HEINLEINIAN WORKSHOP

Korzybski, Twain, Cabell, and the Bootstrap Corpus

Heinlein's own avowed intellectual sources, read against the story, convict its narrator of the very error they were invoked to cure. This chapter treats three documented influences and the author's prior corpus as the workshop in which the 1958 story was assembled.

§ 2.1

General Semantics: Time-Binding and the "Is" of Identity

Alfred Korzybski's *Manhood of Humanity* (1921) classified plants as energy-binders, animals as space-binders, and humans as *time-binders*: beings whose distinctive capacity is to transmit and accumulate knowledge across generations, so that each begins where its predecessors ended. *Science and Sanity* (1933) elaborated a "non-Aristotelian" system whose slogan—the map is not the territory—indicted the subject-predicate grammar of Indo-European languages and, above all, the "is" of identity, which Korzybski held to be false to the structure of a world in which no two things, and no thing at two times, are identical. Among his remedial "extensional devices" was *dating*: Smith-1920 is not Smith-1940.

Heinlein's immersion in General Semantics from the late 1930s is well documented; he expounded time-binding in his Guest of Honor address at the 1941 World Science Fiction Convention in Denver, "The Discovery of the Future," and dramatized Korzybskian themes in "Coventry" (1940) and "Gulf" (1949). A. E. van Vogt's *The World of Null-A* (1945; book 1948), the genre's most conspicuous General Semantics novel, adds a relevant motif: its hero, Gilbert Gosseyn, survives death by awakening in successive bodies, so that personal identity is borne by a series of physically distinct vehicles.

Read with Korzybski, the story is a case study in semantic pathology. Its timestamps are Korzybskian dating devices applied to a person: the infant of 1945, Jane of 1963, the Unmarried Mother of 1970, the bartender of 1970, the narrator of 1993. The narrator's final self-understanding performs the maximal identification Korzybski condemned, collapsing every dated index into a single "I" that is mother, father, child, seducer, and recruiter at once. The metaphysical fact of the story—that these are stages of one continuant—is not in question. What is in question is the semantic posture adopted toward it, and here Heinlein's own master supplies the therapy his narrator lacks: to index the selves, to treat Jane-1963 as structurally other than the bartender-1970, and thereby to recover, within a single worldline, the minimal plurality that the loop otherwise abolishes. Chapter 8 shows that Derek Parfit's reductionism reaches the same counsel by a different road.

The deeper Korzybskian irony concerns time-binding itself. The narrator binds time in the most literal sense available to fiction; yet the loop transmits nothing, since nothing new can enter a closed causal circuit and no generation succeeds another. He is a time-binder who binds no time. The story is a *reductio* of the individualist misreading of Korzybski's anthropology:

time-binding is a property of the species across generations, and a single subject who becomes his own generations does not accumulate but merely recirculates.

§ 2.2

Twain: Determinism and Solipsism

Heinlein, raised in Missouri, repeatedly acknowledged Mark Twain as a formative presence, and two late Twain texts supply the story's premises. *What Is Man?* (1906), a dialogue between an Old Man and a Young Man, argues that a human being is a machine whose every act is determined by temperament and training. *The Mysterious Stranger* (posthumously assembled, 1916) ends with Satan's revelation to the boy Theodor that the universe, God, and the human race are a dream, and that the hearer alone exists: "Nothing exists but you. And you are but a thought—a vagrant thought, a useless thought, a homeless thought, wandering forlorn among the empty eternities!"

The conjunction of these two texts is the logical skeleton of "—All You Zombies—": a deterministic universe (the loop admits no alternative) in which, at the end, the narrator discovers that only he is real. Heinlein transposes Twain's revelation from the register of dream to that of causal topology. What in Twain is the disclosure of an illusion becomes in Heinlein the consequence of a structure; what in Twain is a homeless thought becomes in Heinlein a self-housed worldline. *A Connecticut Yankee in King Arthur's Court* (1889), with its engineer-impresario manipulating a past society, prefigures the bartender as technician of history, though Hank Morgan manipulates the past of others and the bartender only his own.

§ 2.3

Cabell: Irony, Recurrence, and the Sigil

Heinlein's admiration for James Branch Cabell is visible in *Glory Road* (1963) and explicit in *Job: A Comedy of Justice* (1984). Three Cabellian structures bear on the story. *Jurgen* (1919) grants its pawnbroker a recovered youth and a tour of mythic realms, only to show that a second chance repeats the first because the chooser is unchanged. *The Cream of the Jest* (1917) sends Felix Kennaston on dream-voyages through history by means of the Sigil of Scoteia, a half-disc that proves to be the broken lid of a cold-cream jar: the talisman of temporal passage exposed as a cosmetic trifle. And Cabell conceived his eighteen-volume *Biography of the Life of Manuel* as the life of a single man continued through his descendants across centuries.

The ouroboros ring is Heinlein's sigil, a talisman whose meaning is the closed circle of its bearer's life. And the story compresses Cabell's genealogical conceit to its limit: where Cabell treats a lineage of many persons as the extended biography of one, Heinlein makes one person the whole of his lineage. The Cabellian irony—that the hero's repetitions disclose an unchangeable self—is preserved, but its comic register is converted into melancholy.

§ 2.4

The Bootstrap Corpus

Heinlein's first published story, "Life-Line" (*Astounding*, August 1939), already articulates the four-dimensional metaphysics of persons. Its Dr. Pinero, inventor of a device that measures the date of a subject's death, describes the human life as a long worm extending through time from birth to death, whose terminus his instrument detects. Twenty years later, in "—All You Zombies—," that worm bites its tail: the four-dimensional worm of 1939 becomes the Worm Ouroboros of 1959. The trajectory of Heinlein's metaphysics across these two decades can be stated as the bending of a perdurantist worm into a ring.

"By His Bootstraps" (*Astounding*, October 1941, as by Anson MacDonald) gave the genre its name for the information loop. Bob Wilson passes through a Gate, meets several versions of himself in a single room, and becomes Diktor, the ruler of a far-future people rendered docile by vanished masters. Diktor's vocabulary of the future language is copied from a notebook he received from his own younger self, so that the notebook has no author. Two features anticipate 1958: the closed loop of information, which becomes a

closed loop of genealogy; and the docile, will-less population that surrounds the self-ruling self, which becomes the zombies of the title. The pairing of a self-enclosed sovereign with a surrounding populace lacking initiative is thus present in Heinlein seventeen years before it receives its name.

"They" (*Unknown*, April 1941) presents an asylum patient convinced that the world is a stage set maintained by conspirators to deceive him; the ending confirms the conviction. *The Unpleasant Profession of Jonathan Hoag* (*Unknown Worlds*, 1942) presents the world as an artwork under the revision of its critics, its edges dissolving into formless gray. *The Door into Summer* (1957) makes its engineer-protagonist the secret benefactor of his own past through cold sleep and time travel. The later oeuvre returns obsessively to the 1958 matrix: *I Will Fear No Evil* (1970), in which an aged magnate's brain, transplanted into a young woman's body, conceives a child by insemination with his own banked semen; *Time Enough for Love* (1973), in which Lazarus Long travels back to become his own mother's lover and confronts female clones of himself; and the "World as Myth" novels of the 1980s.

§ 2.5

The Solipsist–Pantheist Axis

The corpus describes an axis that the 1958 story hinges. At one pole stands "They" (1941): solipsism confirmed, the world a set built for one spectator. At the other stands *Stranger in a Strange Land* (1961), whose Martian-raised messiah teaches that "Thou art God": pantheism as the plural inversion of solipsism, each person divine. Heinlein's late fiction attempts a reconciliation under the doctrine of "World as Myth," in which every sufficiently vivid imagination generates a universe—what its characters describe as a pantheistic, many-personed solipsism.

"—All You Zombies—" is the last word of the solipsistic phase, written two years before the pantheistic formula. Both texts culminate in a second-person apostrophe: the narrator of 1959 addresses the reader as a zombie; the messiah of 1961 addresses the reader as God. The grammatical form is identical and the valence inverted. Between 1958 and 1961, in other words, Heinlein reverses the sign of "you." The thesis of this dissertation is that the 1958 story already contains the argument for that reversal, encrypted in the self-referential structure of its insult.



THEOLOGIES OF SELF-ORIGINATION

Aseity, Causa Sui, Gnostic Autogenes, and the Unmarried Mother

Every major prohibition that classical theology placed on self-causation returns in the story as a narrative consequence. Heinlein builds, in secular and technological terms, the being that the theological tradition declared either impossible (Aquinas), reserved to God (Anselm, Spinoza), or diabolical (Milton), and he delivers it to precisely the fate that tradition predicted.

§ 3.1

The Classical Prohibition: Aristotle and Aquinas

Aristotle's analysis of self-motion (*Physics* VIII.5) decomposes every apparent self-mover into a moving part and a moved part; nothing moves itself in the same respect in which it is moved. The prime mover of *Metaphysics* Λ is unmoved rather than self-moved. Aquinas's Second Way (*Summa Theologiae* I, q. 2, a. 3) states the corresponding prohibition for efficient causation: nothing can be the efficient cause of itself, for it would then be prior to itself, which is impossible.

The story does not violate the prohibition so much as dissolve its premise. "Prior to itself" is contradictory only on the assumption of a single temporal order. Under Lewis's distinction, Jane is prior to herself in one ordering and posterior in the other: in external time the infant deposited in 1945 precedes the woman seduced in 1963, while in personal time the woman delivering in 1964 precedes the infant she delivers. The Thomistic impossibility, *prius seipso*, becomes a consistent structure, *prius et posterius seipso*, at the price of splitting time into two orders. The question this dissertation presses is what that splitting costs the subject who inhabits it.

§ 3.2

Aseity and Abaliety: Anselm and the Scholastics

Anselm's *Monologion* (1076), chapters 6–7, holds that the supreme nature exists *per se* and *ex se*, through and from itself, yet denies that it is caused by itself. Its self-existence is like the relation of light, to shine, and shining: not an efficient production but an identity of being and act. Scholastic vocabulary crystallized the distinction as *aseitas* (being from oneself) and *abalietas* (being from another), the latter characterizing every creature without exception.

The crucial point is that classical theism ascribes aseity to God precisely by *denying* self-causation: God is uncaused, not self-caused. Heinlein's Jane achieves a counterfeit aseity by the opposite route, through efficient self-causation distributed across her temporal parts. The scholastics would have regarded such "efficient aseity" as a contradiction in terms. The story treats it as a consistent possibility and then explores what kind of being it produces.

§ 3.3

Causa Sui: Descartes and Spinoza

Descartes, answering Caterus in the First Replies and Arnauld in the Fourth, defended a positive sense in which God derives existence "from himself," as quasi-efficient cause of himself, and retreated under pressure to the analogy of a formal cause. Spinoza made the notion definitional: *Per causam sui intelligo id, cujus essentia involvit existentiam; sive id, cujus natura non potest concipi, nisi existens* (*Ethics* I, Def. 1). Only substance satisfies the definition; finite things are modes.

Jane is a mode that has usurped the definition of substance. Within the loop her existence follows from her own nature, in the sense that it follows from her own later acts. The bootstrap is thus a temporal version of the ontological argument: Anselm's *Proslogion* derives the existence of a being from its concept; Heinlein derives the existence of a person from that person's own future. The circle that the ontological argument draws in logical space, the story draws in spacetime.

§ 3.4

The Neoplatonic Circle

Plotinus, in the treatise on the will of the One (*Enneads* VI.8), ventures that the One is in a certain sense its own cause, making itself by its own willing. Proclus systematized the cosmic process as a triad of remaining (*monē*), procession (*proodos*), and reversion (*epistrophē*), holding in the *Elements of Theology* (Prop. 146) that the ends of every divine procession are assimilated to their beginnings, so that reversion describes a circle without beginning or end. Jane's life is a Proclean procession that reverts upon its principle, with one decisive difference: the principle to which it reverts is not a higher hypostasis but itself. Neoplatonic reversion is ascent; Heinlein's is recursion.

§ 3.5

Sethian Autogenes and the Unmarried Mother Sophia

In December 1945—the year in which Jane is deposited at the orphanage—the codices of the Nag Hammadi library were unearthed in Upper Egypt. Among them, the *Apocryphon of John* sets out the Sethian theogony with uncanny relevance. The Invisible Spirit's first thought is Barbelo, the androgynous "Mother-Father" (*metropatōr*). From the divine pleroma proceeds the *Autogenēs*, the Self-Begotten. Lowest in the pleroma, Sophia conceives without the consent of her consort, an unpartnered generation that produces a lion-faced serpent, Yaldabaoth, who, ignorant of what lies above him, proclaims that he alone is God and there is no other.

The structural correspondence is close. The androgynous Mother-Father and the Self-Begotten are mythic prototypes of Jane. Sophia, the unpartnered mother, is the theological original of the byline "The Unmarried Mother." And Yaldabaoth, offspring of unpartnered generation, is the archetypal solipsist demiurge, whose boast that there is no other god matches the narrator's terminal conviction that there is no one present but himself. The narrator thus unites, in one subject, the Gnostic mother who generates without a partner and the Gnostic son who believes himself alone. Hans Jonas's *The Gnostic Religion* (1958) and his earlier essay "Gnosticism and Modern Nihilism" (1952) read this mythology as a figure of the existential self thrown into an alien cosmos, a reading published in the very year of the story's composition.

Two logia of the *Gospel of Thomas*, from the same library, deepen the correspondence. Logion 22 makes entry into the kingdom conditional on making the two into one and the male and female into a single one. Logion 114 has Jesus promise to make Mary male, so that she may become a living spirit. Jane's surgical masculinization and her unification of both sexes in a single worldline literalize both sayings, while the narrator's condition at the end—undivided, complete, and utterly alone—suggests what the Gnostic dream of unity costs when realized without a pleroma.

§ 3.6

Marian Paradox and the Modalist Trinity

Dante's prayer to the Virgin opens with three paradoxes: *Vergine Madre, figlia del tuo figlio* (*Paradiso* XXXIII.1). Mary is the creature who mothers her Creator, and is therefore daughter of her son. Jane is the human subject who mothers herself and is the daughter of her own later male self. Dante's third line names Mary *termine fisso d'etterno consiglio*, the fixed terminus of eternal counsel: predestination as the closure of a divine plan upon a single figure. The 2014 film's title, *Predestination*, rediscovers Dante's term.

The byline "The Unmarried Mother" evokes the virginal conception, and the story's staging of it is unexpectedly exact. In Luke 1:35 the Spirit "overshadows" Mary to effect a conception without a human father. In the story the bartender arranges a conception without any father other than the mother herself: he is the agent who brings the begetter to the bearer and removes the child to its appointed place. The Council of Ephesus (431) proclaimed Mary *Theotokos*, God-bearer; Jane is *autotokos*, self-bearer.

The Nicene confession that the Son is begotten, not made, and *homoousios*—consubstantial—with the Father becomes literal in Jane, whose father is consubstantial with her in the most material sense possible. Yet Jane's structure is not orthodox but heretical. Sabellian modalism taught that Father, Son, and Spirit are successive masks (*prosopa*) of a single divine subject; Tertullian mocked its variant, Patripassianism, by saying that Praxeas had put the Paraclete to flight and crucified the Father (*Adversus Praxean* 1). Jane is a modalist deity: one hypostasis appearing successively as father, mother, child, and the spirit-like agent who orchestrates the conception. The Latin *persona*, as Boethius notes (*Contra Eutychen* III), originally named the theatrical mask; Jane has many *personae* in that sense and, Chapter 12 will argue, no personhood in the metaphysical sense that Trinitarian orthodoxy secured precisely by refusing modalism.

Orthodox Trinitarian theology distinguished the divine persons solely by their relations of origin—paternity, filiation, spiration—so that, in Aquinas's formula, a divine person signifies a relation as subsisting (ST I, q. 29, a. 4). The persons are really distinct because their relations of origin are really opposed. Jane's loop counterfeits relations of origin without real distinction: her "father" and her "mother" are not opposed relations but stages of one continuant. The story therefore constitutes a narrative refutation of modalism, demonstrating that where there is no real distinction there is no relation, and where there is no relation there is no person.

One further constellation is too precise to omit. The first atomic explosion, in July 1945, was code-named *Trinity*; J. Robert Oppenheimer later attributed the name to John Donne's Holy Sonnet XIV, which begins "Batter my heart, three-person'd God" and ends in the paradox of enforced liberty: "Except you enthrall me, never shall be free, / Nor ever chaste, except you ravish me." Donne's soul is ravished by a God of three persons; Jane is seduced by the future self who is her father, her child's father, and her own abductor. The ravishing deity of devotional poetry becomes, in Heinlein, the self.

§ 3.7

Milton's Satan and the Infernal Genealogy

In Book V of *Paradise Lost*, Satan answers Abdiel's reminder that the angels were created by denying any memory of creation and claiming that the angels are "self-begot, self-rais'd / By our own quick'ning power" (V.860–861). Milton presents autogenesis as the founding lie of rebellion, the creature's claim to aseity. In Book II, Sin recounts her birth from Satan's head and their union: "Thy self in me thy perfect image viewing / Becam'st enamour'd" (II.764–765), a narcissistic incest whose offspring is Death. And in Book IV Satan confesses the terminus of autogenetic pride: "Which way I fly is Hell; myself am Hell" (IV.75).

Heinlein makes Satan's lie literally true for a human being. The narrator is self-begot; his self-generation is a union with his own image; and its product is not a new being but the same one, a sterile repetition that Milton would have recognized as Death. His terminal condition—alone, with no one present but himself—is Milton's "myself am Hell" rendered as cosmology. Sartre's celebrated identification of hell with other people is thereby inverted: for the narrator, hell is the absence of others, which is to say, himself.

§ 3.8

Nietzsche's Münchhausen and Sartre's Useless Passion

In *Beyond Good and Evil* §21, Nietzsche calls the *causa sui* the best self-contradiction yet conceived and likens the desire for absolute freedom of will to a feat surpassing Münchhausen's: pulling oneself up by the hair out of the swamp of nothingness into existence. The English idiom of "bootstraps" descends from the same comic tradition, and Hans Albert's "Münchhausen trilemma" (*Traktat über kritische Vernunft*, 1968) named the three failures of foundationalism after it: infinite regress, dogmatic termination, and circularity. Jane is the circular horn of the trilemma made flesh.

Sartre's *L'Être et le néant* (1943) diagnosed human reality as the desire to be God, understood as the impossible synthesis of the in-itself and the for-itself, the *ens causa sui*; the project is necessarily futile, and the human being a useless passion. Heinlein's narrator achieves what Sartre declared unachievable and discovers that its achievement is not plenitude but desolation. The story thus vindicates Sartre *a contrario*: even if the passion were not useless, its fulfillment would be uninhabitable.

§ 3.9

Synthesis: Technological Aseity

Darko Suvin defined science fiction by its *novum*, a cognitively validated innovation that estranges the empirical world. The novum of "—All You Zombies—" is not the time machine as such but its function: it is a machine for producing aseity. The Temporal Bureau, which deploys it, is the apparatus of a secular providence that arranges a creature's origin from itself. Theology prohibited self-causation as impossible, reserved it to God, or condemned it as satanic. The story demonstrates that, once made technically possible, self-causation inherits all three verdicts at once: it is possible only through a split time, divine only in its formal structure, and satanic in its consequence.



ICONOGRAPHIES OF CLOSURE

Ouroboros, Androgyne, Narcissus, Tiresias

The story's single emblem, the ouroboros ring, condenses four iconographic traditions, and each tradition, followed to its source, already contains the verdict the story reaches: closure upon oneself is figured throughout Western myth as plenitude in form and impoverishment in substance.

§ 4.1

The Serpent That Devours Itself

The earliest known ouroboroi encircle a figure on the second gilded shrine of Tutankhamun, in the *Enigmatic Book of the Netherworld*, where serpents biting their tails surround head and feet in a figure of solar regeneration. Greek alchemy inherited the image: the *Chrysopoeia* of Cleopatra, preserved in a Byzantine manuscript in Venice, inscribes the ouroboros with the formula *hen to pan*, "the One is the All." Horapollo's *Hieroglyphica* (I.2) reads the tail-eating serpent as the cosmos. Heraclitus had already supplied the geometric principle: on the circumference of a circle, beginning and end are common (DK B103). Norse myth encircles Midgard with Jörmungandr, its tail in its mouth; the Gnostic *Pistis Sophia* figures the outer darkness as a great dragon with its tail in its mouth; Kekulé, recounting in 1890 the reverie that disclosed the benzene ring, described a snake seizing its own tail.

The formula *hen to pan* is the story in two words: the one person who is all the persons of his genealogy. E. R. Eddison's *The Worm Ouroboros* (1922), whose title the narrator's ring invokes, ends by looping to its beginning: the victors, bored by peace, ask the gods to restore their slain enemies so that the war may start again. Eddison's circularity is aristocratic ennui; Heinlein's is ontological necessity. A further sense of the ring has been overlooked. A ring worn on the hand is conventionally a token of marriage. The Unmarried Mother is, in the end, married—to himself—and the ring is the sacrament of that union.

§ 4.2

The Androgyne: Aristophanes, the Rebis, and the Myth of Wholeness

In Plato's *Symposium* (189c–193d), Aristophanes tells of the original humans: spherical beings of three sexes—male, female, and androgynous—cut in half by Zeus for their arrogance, each half thereafter seeking its other. Eros is the desire for this lost wholeness. Hephaestus, Aristophanes imagines, might offer to weld lovers into one being forever (192d–e).

Jane realizes Hephaestus's offer. Her female and male halves are literally one being, welded not in space but along a worldline. The result refutes Aristophanes. The narrator, complete, still longs—he confesses that he misses the girl he was—so that eros survives the attainment of wholeness. What eros sought was therefore not completion but alterity, the other as other.

Diotima's correction, a few pages later in the same dialogue (205e), anticipates the verdict: love is of neither the half nor the whole, unless that half or whole is good.

The alchemical tradition gave the androgyne a chemical body, the *Rebis* or "double thing," depicted in the *Rosarium philosophorum* (1550) as the royal hermaphrodite produced by the *coniunctio* of Sol and Luna, frequently figured as the incestuous union of brother and sister. Jung's *Psychology and Alchemy* (1944), "The Psychology of the Transference" (1946), and *Mysterium Coniunctionis* (1955–56) read the coniunctio as an image of the Self, the union of opposites that Nicholas of Cusa had attributed to God as *coincidentia oppositorum* (*De docta ignorantia*, 1440). Erich Neumann's *The Origins and History of Consciousness* (1949) placed the uroboros at the beginning of psychic development, a pre-egoic Great Round, and named the ego's regressive dissolution into the maternal origin "uroboric incest." Jane's self-union is uroboric incest literalized: not the integration of opposites that Jung prized but a regression that wears the mask of completion.

§ 4.3

Autochthony and the Structural Study of Myth

Athenian civic myth claimed that the citizens were *autochthonous*, born from the earth; their ancestor Erichthonius sprang from the soil where Hephaestus's seed had fallen. Nicole Loraux's *Les Enfants d'Athéna* (1981) and *Né de la terre* (1996) read autochthony as a political fantasy of reproduction without women. Plato's *Statesman* (269c–274e) imagines a reversed cosmic era in which humans are born old from the earth and grow young.

Claude Lévi-Strauss's "The Structural Study of Myth" (*Journal of American Folklore*, 1955), published three years before the story, analyzes the Oedipus myth as an attempt to mediate a contradiction between the belief that humans are autochthonous—born from one—and the knowledge that each is born of a man and a woman—born from two. The myth asks, in his reformulation, whether the same is born from the same or from the other. And myth, he argued, provides a logical model capable of overcoming a contradiction, an impossible achievement if the contradiction is real.

Jane is a Lévi-Straussian mediator of extraordinary purity: born from two (a mother and a father) who are one (a single person). The story answers the mythic question with both terms at once: the same is born from the same, by way of the other sex of the same. If Lévi-Strauss is right that myth cannot

dissolve a real contradiction, the residue of the failed mediation should appear somewhere in the narrative. It does: in the narrator's terminal solitude, the remainder that no logical model can absorb.

§ 4.4

Ovid's Book III: Tiresias and Narcissus

The third book of the *Metamorphoses* places side by side the two figures whose synthesis Jane is. Tiresias, having struck two copulating serpents, lives seven years as a woman before regaining his male form (III.316–338); called to arbitrate whether men or women derive greater pleasure from love, he is blinded by Juno and compensated by Jupiter with prophecy. His first prophecy concerns Narcissus. Asked whether the boy will live long, Tiresias answers: *si se non noverit*, if he never knows himself (III.348).

The inversion of the Delphic precept is the hinge of the story. Narcissus, at the pool, recognizes his beloved as his own reflection: *iste ego sum: sensi, nec me mea fallit imago; / uror amore mei* (III.463–464), "I am he; I have felt it, my image does not deceive me; I burn with love of myself." Then comes the line that serves as this dissertation's second epigraph: *quod cupio mecum est: inopem me copia fecit* (III.466), "what I desire is with me; my abundance has made me poor."

Jane is a Tiresian Narcissus. Like Tiresias, she has lived as both sexes and, as the bartender, possesses foreknowledge. Like Narcissus, she desires and possesses herself, and comes to know herself in the most literal manner conceivable, by meeting herself. Tiresias's prophecy is fulfilled: the subject who knows herself completely cannot live, in the full sense, among others. The narrator's final condition, possessing all of himself and impoverished by that possession, is *inopem me copia fecit* in cosmological form. Echo, condemned to repeat the last words of others, completes the Ovidian triad: the bartender's foreknowledge is an echo of his own past, and the narrator, like Echo, fades at the end into a voice in the dark.

A fourth book supplies the corporeal model. Salmacis fuses with Hermaphroditus into a single body that is *neutrumque et utrumque*, neither and both (IV.379). Ovid's fusion is simultaneous; Heinlein's is successive. Jane is each sex in turn, and both only across time.

§ 4.5

Eliot's Tiresias: The Foresuffering Witness

The Waste Land (1922) gives Ovid's prophet his modernist form: "I Tiresias, though blind, throbbing between two lives, / Old man with wrinkled female breasts, can see" (ll. 218–219). Eliot's note to these lines is the most precise gloss on Heinlein's narrator ever written, though written thirty-six years before him: Tiresias, "although a mere spectator and not indeed a 'character,' is yet the most important personage in the poem, uniting all the rest … so all the women are one woman, and the two sexes meet in Tiresias. What Tiresias sees, in fact, is the substance of the poem."

The bartender is a Tiresian narrator in every term of the note: a spectator rather than a protagonist, in whom the two sexes meet, uniting all the other characters, whose vision is the substance of the narrative. The correspondence extends to the action. Eliot's Tiresias witnesses the loveless seduction of a typist by a clerk, and interjects: "And I Tiresias have foresuffered all / Enacted on this same divan or bed; / I who have sat by Thebes below the wall / And walked among the lowest of the dead" (ll. 243–246). The bartender has, in exactly this sense, foresuffered the seduction he arranges: he was its object before he became its orchestrator. And Tiresias has walked among the lowest of the dead.

The Eliotic genealogy of the title follows. Earlier in the poem, the crowd flowing over London Bridge is beheld as a procession of the undead: "Unreal City, / Under the brown fog of a winter dawn, / A crowd flowed over London Bridge, so many, / I had not thought death had undone so many" (ll. 60–63), translating Dante's vision of the souls in the vestibule of Hell (*Inferno* III.55–57). The modern crowd as walking dead, beheld by a solitary witness in whom all persons are one person: this is the scene of Heinlein's final apostrophe. The story is, in effect, a science-fictional transposition of *The Waste Land*'s Tiresian structure into causal topology. What in Eliot is an identity of perspective—"the two sexes meet"—becomes in Heinlein an identity of persons.

§ 4.6

Woolf's Orlando and the Androgynous Mind

Virginia Woolf's *Orlando* (1928) follows a single person across four centuries and a change of sex, received with serene equanimity, culminating in the completion of a long-deferred poem, a marriage, and an immersion in the life of a historical society. *A Room of One's Own* (1929) commends, after Coleridge's remark that a great mind is androgynous, the creative fusion of masculine and feminine in a single sensibility. Woolf's androgyny is sociable, generative, and historically embedded; it opens outward. Heinlein's is solitary and closed. Orlando acquires a room of her own and fills it with others; Jane acquires something stranger, a womb of her own, the most solitary chamber imaginable, since what it bears is the bearer.



FORM AS TOPOLOGY

Timestamp, Confession, and the Autodiegetic Loop

The story's narrative form does not merely represent its metaphysics; it enforces it. Every addressee inside the diegesis is the narrator himself, so that the only genuine other the text admits is the reader outside it, and that reader is the one the narrator names a zombie.

§ 5.1

Fabula and Syuzhet in a Closed Timelike Narrative

In ordinary narrative the reconstruction of the *fabula* from the *syuzhet* yields a line, however scrambled its presentation. Here it yields the σ-topology described in the Introduction: a loop with a tail. The *syuzhet* follows the narrator's personal time, segment by segment, while the reader reconstructs an external-time *fabula* that bends back upon itself. Wittenberg (2013) has argued that time-travel fiction is a "narratological laboratory" in which the ordinary prerogatives of the narrator—to move freely among the times of the story—are literalized as the prerogatives of a character. Heinlein's narrator is a limit case: his freedom of narrative movement coincides exactly with his causal movement, and both are confined to his own life.

§ 5.2

The Timestamp as Chronotope

Mikhail Bakhtin's "Forms of Time and of the Chronotope in the Novel" (written 1937–38) defined the chronotope as the intrinsic connectedness of temporal and spatial relationships artistically expressed in literature. Each genre has its characteristic chronotopes: the road, the castle, the provincial town, the threshold. The chronotope of "—All You Zombies—" is the *coordinate*: a time, a time-zone designation, a date, and a place, set as a heading over each segment. This is a bureaucratic chronotope, descended from the naval log. Heinlein, a 1929 graduate of the U.S. Naval Academy who served until his medical retirement in 1934, gives his agent the documentary habits of a watch officer.

The coordinate chronotope has three functions. Physically, each heading designates an event in Minkowski spacetime, so that the narrative proceeds literally from event to event along a worldline. Semantically, each heading is a Korzybskian date applied to a person. Metaphysically, each heading inscribes McTaggart's B-series over a narration conducted in the A-series. Bakhtin observed that the threshold chronotope is the time of crisis and decision. The coordinate is its antithesis: it records events as already fixed, so that no segment of the story is a threshold. Nothing is decided in the story; everything is executed.

§ 5.3

The Confession: Pulp Genre and Gender Masquerade

The Unmarried Mother writes for the confession magazines, a pulp genre inaugurated by Bernarr Macfadden's *True Story* (1919), whose first-person narratives of transgression, suffering, and redemption followed a strict formula and were not infrequently produced by male professionals writing as women. The byline is thus a textual intersex. Before and after the surgical reassignment, the writer's public voice remains that of a woman, so that the bodily transformation is bracketed by a persistent feminine persona in print.

The frame turns on a wager about the singularity of a life story: the young man bets that his history is stranger than any the bartender has heard. The confession genre lives on the claim that each life is unique and survives by the repetition of a formula. The story stages this paradox of genre as a paradox of identity. Every confession is the same confession because every confessor is the same confessor; the bartender cannot be told a story new to him because the story is his own. The terms of the wager are undecidable from within the frame, since the auditor is the life in question.

§ 5.4

The Augustinian Armature

The scene of confession to an auditor who already knows the story has a canonical original. Book XI of Augustine's *Confessions* opens by asking why he narrates his life to a God who is eternal and therefore cannot be ignorant of what he is told: *Numquid, domine, cum tua sit aeternitas, ignoras quae tibi dico, aut ad tempus vides quod fit in tempore? Cur ergo tibi tot rerum narrationes digero?* (XI.1.1). Augustine answers that he narrates not to inform but to stir up love, in himself and in his readers.

The bartender is Augustine's eternal auditor rendered as a person. He exists outside the young man's time in the sense that he has lived beyond it; he knows every event of the narrative because he lived it. His purpose in hearing the confession is likewise not to learn but to produce. The confession arouses the young man's desire for revenge, and that desire drives him into the seduction that closes the loop. Confession here is not a report on a life but an instrument of its causation.

The macrostructure of the *Confessions* sharpens the correspondence. Books I–IX narrate a life; Book X turns to memory; Book XI to time, including the celebrated aporia *Quid est ergo tempus? Si nemo ex me quaerat, scio; si*

quaerenti explicare velim, *nescio* (XI.14.17); Books XII–XIII to creation, through an exegesis of Genesis. The story traverses the same sequence in compressed form: the young man's life story; the bartender's memory of it; the time machine; and the closing question of origins. The difference is directional. Augustine's narrative moves toward God as the source of the self; Heinlein's moves toward the self as the source of the self. The *Confessions* open with the heart's restlessness until it rests in God (*inquietum est cor nostrum*, *donec requiescat in te*, I.1.1). Heinlein's narrator has a heart that can rest only in itself, and therefore cannot rest.

§ 5.5

Narrative Levels and the Autonarratee

Gérard Genette's *Discours du récit* (1972) distinguishes narrative levels (extradiegetic, intradiegetic, metadiegetic) and relations of narrator to story (heterodiegetic, homodiegetic, and the strong form of the latter, autodiegetic, in which the narrator is the protagonist). In the story the frame narrator, the bartender, is autodiegetic. The embedded narrator, the young man telling his confession, is likewise autodiegetic. The narratee of the embedded tale is the frame narrator. And the narrator of the embedded tale is the frame narrator's past self.

Narrator and narratee are therefore identical across levels. Genette's typology has no term for this, and this dissertation proposes one: the **autonarratee**, a narratee identical with the narrator in a relation that crosses diegetic levels. Every speech act inside the story is addressed by the self to the self. The seduction scene, the recruitment, and the confession are all exchanges within a single person.

The closing apostrophe is therefore a metalepsis in the strict sense: a transgression of narrative levels in which the narrator addresses the extradiegetic narratee, the reader. That reader is the only addressee in the entire text who is not the narrator. The reader is thus the only other the story contains, and it is precisely this other whom the narrator denies personhood. Louis Althusser's "Idéologie et appareils idéologiques d'État" (1970) described interpellation as the hailing by which individuals are constituted as subjects. The apostrophe performs a *dis-interpellation*: a hailing that constitutes its addressee as a non-subject. The structure of narration thus enforces the solipsism of the content. A narrator whose every interlocutor is

himself can meet the other only at the boundary of the text, and can greet him there only with negation.

§ 5.6

The Knot That Unties in Four Dimensions

Viewed in spacetime, the narrator's worldline is a self-touching curve with several events of co-presence: Cleveland 1963 (Jane, the seducer, and nearby the bartender), Cleveland 1964 (the mother, the infant, and the abductor), New York 1970 (the young man and the bartender). Projected onto a diagram of three dimensions, it presents itself as a knot.

A theorem of geometric topology supplies a figure for the reading. Every knotted circle in three-dimensional space can be unknotted in four dimensions: there are no nontrivial one-dimensional knots in four-space. The "paradox" of the story is, in this figure, a projection artifact of the tensed, three-dimensional perspective of a reader who experiences the narrative as a succession; from the standpoint of the four-dimensional block, the worldline is merely a curve. Yet persons do not inhabit the block; they live in perspectival, tensed time. The tension between the untangled curve of the block and the lived knot of the person is the story's form.

Two further figures complete the topology. August Möbius and Johann Listing described the one-sided band in 1858, a century before the story. A figure transported once around a Möbius band returns mirror-reversed. The paternal circuit of Jane's worldline—from the infant through Jane to the young man who returns to 1963 as her seducer—is orientation-reversing in exactly this sense, since the subject re-enters the event of conception with sex inverted. And Guillaume de Machaut's rondeau *Ma fin est mon commencement* is a fourteenth-century retrograde canon whose voices read the same music forwards and backwards. The story read backward from its coda discloses the same person at every point. Its end is its beginning, sung in retrograde.



THE IMPOSSIBLE BODY

Sex, Surgery, Reproduction, and the Biopolitics of the Loop

Physics permits the narrator's loop; biology forbids his body. This chapter shows that the story's metaphysical possibility is purchased at the price of a biological impossibility, and that the impossible body is nonetheless saturated with the real medical, sexual, and institutional history of mid-century America.

§ 6.1

Intersex Medicine at Mid-Century

The surgeons who reconstitute Jane act within a recognizable medical regime. Hugh Hampton Young's *Genital Abnormalities, Hermaphroditism, and Related Adrenal Diseases* (1937) had codified the surgical management of intersex conditions. John Money and Joan and John Hampson's papers of 1955 in the *Bulletin of the Johns Hopkins Hospital* introduced the concept of "gender role" and the protocol of early surgical assignment followed by consistent rearing. Christine Jorgensen's return from Copenhagen in December 1952 had made sex reassignment front-page news. Behind these stood Magnus Hirschfeld's Institut für Sexualwissenschaft (1919–1933), sacked on 6 May 1933 by Nazi students of the Deutsche Studentenschaft, followed by SA men, its library burned in the Berlin book burning of 10 May; and the operations on Lili Elbe (1930–31), the last of which, an attempted uterus transplantation performed in Dresden by the gynecologist Kurt Warnekros, was followed by her death in September 1931. The story's surgeons decide during the Caesarean, while Jane is unconscious, to remove her female organs and to begin reconstituting her as a man, and they tell her only afterward; the decision is theirs alone. This is the medical paternalism against which intersex activism, organized from 1993 in the Intersex Society of North America, would later protest.

The record of that regime names its agents. On John Money's advice, physicians at Johns Hopkins surgically reassigned David Reimer as a girl in 1967, after a botched circumcision in infancy, and treated him with estrogen at puberty; Money reported the case for years as evidence that rearing determines gender identity (Money and Ehrhardt 1972). Milton Diamond and Keith Sigmundson showed in 1997 that Reimer had rejected the assignment in adolescence and was living as a man. Reimer spoke publicly against the treatment and died by suicide in 2004; how far the treatment contributed is disputed. Intersex adults, organized in the Intersex Society of North America by its founder, Cheryl Chase, testified that genital surgeries performed on them in infancy and childhood, without their consent and often without disclosure, caused lasting physical and psychological harm. The United Nations Special Rapporteur on torture condemned such non-consensual surgeries in 2013, and Human Rights Watch and interACT documented their continuation in the United States in 2017. Jane's surgeons belong to this regime: they decide the sex of a patient who cannot speak, on the theory that sex is a matter for medical assignment. Lili Elbe's last operation supplies the inverse

case. She sought, at mortal risk, the capacity to gestate that Jane's surgeons remove without asking her.

§ 6.2

The One-Sex Model and the Teleology of Perfection

Thomas Laqueur's *Making Sex* (1990) reconstructed the premodern "one-sex model," in which female genitalia were understood, after Galen, as male genitalia inverted and retained within the body. Aristotle had called the female a kind of mutilated male (*Generation of Animals* II.3, 737a). Renaissance medicine drew a corollary. Ambroise Paré's *Des monstres et prodiges* (1573) and Montaigne's essay on the force of imagination (*Essais* I.21) report cases of women transformed into men—Montaigne's Marie Germain, whose male parts descended as she leapt a ditch—while insisting that the reverse never occurs, because nature always tends toward the more perfect.

Jane's trajectory from female to male re-enacts this early-modern teleology under modern surgery, and the story, by its narrator's own account of his new life, appears at first to endorse it: the male stage is the stage of agency, travel, and command. But the coda reverses the moral. The narrator's confession that he misses the girl he was undermines the teleology of perfection from within. The story is a Renaissance prodigy narrative restaged with its valuation inverted: the "perfection" of maleness is experienced as loss.

§ 6.3

The Genomic Fixed Point

The story's biology can be stated with formal precision. Let G denote Jane's diploid genome. Sexual reproduction is a stochastic map R taking a maternal and a paternal genome to a filial one, through meiosis, recombination, fertilization, and mutation. In Jane's case mother and father are the same organism, so the filial genome is $R(G, G)$, the product of self-fertilization. Novikov consistency requires that the filial genome be Jane's own, since the child is Jane:

$$R(G, G) = G$$

The genome must be a fixed point of the selfing map. Four consequences follow.

1. **Heterozygosity.** At any locus where Jane carries two different alleles, self-fertilization yields a heterozygote with probability one half. With linkage, exact reconstitution of both parental haplotypes further requires that the two gametes be precise complements at every crossover. For a human genome carrying millions of heterozygous sites and several dozen crossovers per meiosis, the probability of exact reconstitution is effectively nil.
2. **Homozygosity.** Consistency becomes probable only if G is homozygous at every locus, so that any selfed gamete reproduces it. In Sewall Wright's terms (*American Naturalist*, 1922), Jane's coefficient of inbreeding must equal unity. A single generation of selfing from an outbred parent yields a coefficient of one half; a coefficient of one is the asymptote of indefinitely repeated selfing. Jane's genome must therefore bear the signature of an inbreeding history that never occurred.
3. **Mutation.** The human germline introduces on the order of sixty to seventy new point mutations per generation. Consistency requires that none occur in Jane's conception. Under a Poisson model the probability of zero new mutations is roughly e^−60 to e^−70, between about 10^−26 and 10^−31.
4. **Sex determination.** Functional ovotesticular development capable of both spermatogenesis and oogenesis is essentially unattested in humans. A 46,XX/46,XY chimera, the most plausible substrate, arises from the fusion of two zygotes and is not heritable; its recurrence in the child would be a further independent coincidence.

Paul Horwich (1987) argued that closed timelike curves, though consistent, require improbable coincidences; Nicholas Smith (1997) replied that, conditional on the loop's existence, its consistency has probability one. The story embodies the dispute in flesh. Jane's body is the coincidence made organic: prior to the loop, astronomically improbable; given the loop, necessary.

The interpretive conclusion is a **chronobiological antinomy**. General relativity tolerates closed timelike curves (Gödel, van Stockum, Morris–Thorne–Yurtsever), and the Novikov principle tolerates consistent loops through them. Human reproductive biology does not tolerate a self-identical selfing. The story's metaphysical possibility rests on a biological impossibility, and the impossible body signals that autogenesis is representable only at the price of fantasy. Deutsch's knowledge paradox (§1.7) ap-

plies with full force: Jane's genome is information without origin. The notebook of "By His Bootstraps" has become a genome.

Two biological contrasts frame the antinomy. Aldous Huxley's *Brave New World* (1932) imagined the Bokanovsky Process, which divides one fertilized egg into as many as ninety-six identical twins: many from one. Heinlein imagines one from one. The two are complementary limits of reproductive control: mass replication and singular self-replication. And evolutionary biology explains sexual reproduction by its production of difference. Leigh Van Valen's Red Queen hypothesis (1973) and H. J. Muller's ratchet (1964) account for the persistence of sex by the advantages of recombination and the degeneration of asexual lineages. Jane's reproduction has the form of sex and the content of cloning. It is sex without difference, retaining the morphology of sexual reproduction while cancelling its evolutionary function.

§ 6.4

W.E.N.C.H.E.S.: The State, Sexual Labor, and Space

The corps Jane aspires to join—Women's Emergency National Corps, Hospitality & Entertainment Section—parodies the U.S. Navy's WAVES (Women Accepted for Volunteer Emergency Service, established 1942), in which Heinlein's third wife, Virginia, served. It reflects Heinlein's sexual libertarianism and its contradictions. The state instrumentalizes women's sexual labor in the service of male spacemen, demanding sexual availability without reproduction. Leiber's *The Big Time* (1958), with its recuperative "Entertainers" serving the soldiers of the Change War, supplies a contemporaneous parallel.

Michel Foucault's *La Volonté de savoir* (1976) described the modern state's deployment of sexuality and management of reproduction as biopower. In the story the biopolitical contradiction is exact. Jane's ambition to reach space—the future—is channeled into sanctioned sexual service; pregnancy expels her from the program and hence from the future; and the future reopens only to her masculinized self, recruited by the Temporal Bureau as a man.

§ 6.5

The Unwed Mother and the Baby Scoop Era

Between roughly 1945 and 1973, unwed mothers in the United States, most of them white, were sent by their families into maternity homes and pressed by caseworkers and hospital staff into relinquishing their infants for adoption, frequently without informed consent; infants were often removed at birth. Rickie Solinger's *Wake Up Little Susie* (1992) and Ann Fessler's *The Girls Who Went Away* (2006) document the practice. Adoption historians have named the period the "Baby Scoop Era." Elaine Tyler May's *Homeward Bound* (1988) placed it within the Cold War ideology of domestic containment.

The agents of the practice can be named. Parents arranged confinements to conceal pregnancies. Maternity homes run by the Florence Crittenton association, the Salvation Army, Catholic Charities, and other religious bodies housed the women and routed their infants to adoption agencies. Caseworkers applied a psychiatric model, codified in Leontine Young's *Out of Wedlock* (1954), that read single pregnancy as a neurotic symptom and surrender as its cure. In hospitals, many women were sedated in labor, kept from seeing or holding their infants, and asked to sign surrender papers while medicated or under pressure; sealed-records laws then cut off contact. The treatment divided by race, which is the central finding of Solinger's study. White single mothers were channeled toward surrender, their infants placed with married white couples; Black single mothers were largely excluded from maternity homes and adoption services and were targeted instead by punitive welfare rules, as when Louisiana's 1960 "suitable home" law removed thousands of children, most of them Black, from public assistance (Bell 1965). Australia's federal government formally apologized in 2013 for forced adoptions carried out in the same decades.

The story's dates fall at the era's boundaries and center. Jane is a foundling in 1945, the era's opening; she is an unwed mother whose infant disappears from the hospital nursery in 1964, its midpoint. The Temporal Bureau's agent performs the role of the institution: he removes the child from an unmarried mother and places her in an orphanage, ostensibly for the good of a larger order. The story literalizes the era's sanctioned abduction, with one decisive turn: the bereaved mother and the bereaving agency are the same person. The narrator administers the biopolitics of the Cold War family to himself. The Bureau is legible as an allegory of the adoption system and the loop as the closed circuit of institutional reproduction.

The Bureau's office begins before the abduction. By the story's dates—deposited at the orphanage on 20 September 1945 at one month old—Jane is seventeen in April 1963, when the bartender conveys a man of twenty-five to her city and, once the conception the loop requires has taken place, withdraws him; what Jane experiences as abandonment is the Bureau's retrieval of its recruit. The seducer knows her loneliness, history, and future from the inside, and she knows nothing of him. The institution does not merely take the child of an unmarried mother. It manufactures the unmarried mother whose child it will take, using a minor's pregnancy to produce its own future agent. Autogenetic Hypostasis (Ch. 12.6) here shows its cost in a single case: a self-founding apparatus that reproduces itself through the body of a girl who cannot know what is being done to her.

§ 6.6

Learning to Be a Man: Performativity Before Performativity

After surgery Jane must learn to inhabit masculinity, acquiring its comportment as a second language. Simone de Beauvoir's *Le Deuxième Sexe* (1949) had argued that one is not born but becomes a woman. Judith Butler's *Gender Trouble* (1990) would theorize gender as performative, a stylized repetition of acts. Between them stands a case almost exactly contemporaneous with the story. In the autumn of 1958, a nineteen-year-old called Agnes presented at the UCLA Medical Center as an intersex person seeking surgery. Harold Garfinkel's study of her, published in *Studies in Ethnomethodology* (1967), analyzed "passing" as the managed achievement of sex status, a continuous practical accomplishment. The Unmarried Mother is such a practitioner, and his career writing confessions as a woman is the continuation of a feminine performance in print after its bodily cessation.

The case had a sequel, which Garfinkel recorded in an appendix. In 1966 Agnes told the psychiatrist Robert Stoller that she had been assigned male at birth and had taken her mother's prescribed estrogen since about the age of twelve (Stoller 1968). She was not intersex. The diagnosis that secured her surgery rested on bodily changes she had induced, at a time when such surgery was granted to patients classified as intersex and refused to those understood as transsexual. Agnes had passed not only as a woman but as a medical category, and the accomplishment Garfinkel analyzed extended into the clinic itself. The sequel sharpens the reading of Jane: the regime that re-

assigned Jane without asking her is the regime that a person seeking reassignment had to deceive in order to obtain it.

§ 6.7

Feminist and Queer Critique: Irigaray, Russ, and the Abolition of Sexual Difference

Joanna Russ's essays "The Image of Women in Science Fiction" (1970) and "What Can a Heroine Do? or Why Women Can't Write" (1972) indicted the genre's women as projections of male need, and Heinlein's women have long been read as competent yet sexually instrumentalized; Mendlesohn (2019) complicates the verdict without dismissing it. Luce Irigaray's *Speculum de l'autre femme* (1974) diagnosed Western philosophy as an economy of the Same in which woman figures only as the specular other of the masculine subject. Freud's lecture "Femininity" (1933) had described the little girl as, in effect, a little man before her discovery of castration.

The story literalizes Irigaray's diagnosis with unsettling exactness: the woman becomes, quite literally, the masculine subject's past stage. Irigaray's *Éthique de la différence sexuelle* (1984) proposed that sexual difference is one of the major philosophical questions, if not the question, of the age. Heinlein's story is perhaps the strongest literary evidence for her claim that sexual difference is the minimal alterity of the human. When the female becomes the male's prior stage, sexual difference is abolished, and with it the last other. There remains only the Same, and the Same is alone.

This reading must coexist with a legitimate reparative one. Recent intersex and trans readings reclaim Jane's trajectory as embodied self-authorship, and nothing in the present argument contests the dignity of gender self-determination. The distinction that reconciles the two readings is between *gender self-determination*, which presupposes a community of others who recognize the self so determined, and *genealogical self-origination*, which forecloses them. The former is a practice of freedom among persons; the latter dissolves the persons among whom freedom would be practiced. The comparative constellation in Chapter 11—Sturgeon's *Venus Plus X* (1960), Le Guin's *The Left Hand of Darkness* (1969), Russ's *The Female Man* (1975), Gerrold's *The Man Who Folded Himself* (1973)—shows how the same materials yield communal rather than solipsistic androgyny.



PSYCHOANALYSIS OF CLOSURE

Family Romance, Primal Scene, and the Inverted Schreber

The story realizes, without remainder, the fantasies that psychoanalysis identified as the unconscious matrix of the family, and the structure that results is not neurotic but psychotic. Its closest psychoanalytic homologue is not Oedipus but Daniel Paul Schreber, inverted.

§ 7.1

The Foundling's Family Romance

Freud's "Der Familienroman der Neurotiker" (1909) described the child's fantasy of being a foundling or stepchild whose true parents are of exalted rank. Jane is a foundling in fact. When her true parents are revealed, they are not exalted but identical with herself, so that the family romance is fulfilled as deflation: the noble origin the child imagines turns out to be the child. Marthe Robert's *Roman des origines et origines du roman* (1972) derived two novelistic types from Freud's essay, the Foundling, who repudiates the family wholesale, and the Bastard, who repudiates the father while acknowledging the mother. Jane is both: a foundling who is also the illegitimate child of a seducer. Robert held that the novel itself originates in the family romance. On her terms, the story is a novel's origin folded into its own conclusion.

§ 7.2

The Rescue Fantasy and the Wish to Be One's Own Father

In "Über einen besonderen Typus der Objektwahl beim Manne" (1910), Freud analyzed the rescue fantasy, in which the son imagines repaying the mother by giving her a child, and found that tenderness, gratitude, lust, defiance, and the claim to independence converge in the single wish to be one's own father. The story grants the wish literally. What follows the fulfillment is instructive: not the cessation of desire but a residual longing for a lost object, the girl the narrator was. The fulfilled wish leaves its subject not satisfied but bereaved.

§ 7.3

The Primal Scene and Nachträglichkeit

In the case of the Wolf Man, "Aus der Geschichte einer infantilen Neurose" (1918), Freud reconstructed a primal scene—the child's observation of parental intercourse—whose traumatic meaning was conferred only later, through *Nachträglichkeit*, deferred action. Constance Penley (1986) identified in time-travel cinema, paradigmatically *The Terminator*, a fantasy of being present at, and causally involved in, one's own conception. "—All You Zombies—" is the limit case of Penley's structure: the subject is not merely present at the primal scene but is both of its participants and its product.

A deeper observation follows. *Nachträglichkeit* is the psychoanalytic name for retrocausation of meaning: a later event determines the significance of

an earlier one. Freud's model of the psyche already contains backward causation at the level of signification. The story physicalizes the principle. Jane's seduction acquires its full meaning only when she returns to it as the seducer; but in the story the later event does not merely reinterpret the earlier, it causes it. Jean Laplanche's *après-coup* becomes efficient causation.

§ 7.4

The Double and the Uncanny

Otto Rank's *Der Doppelgänger* (1914) traced the double from an insurance against death to its harbinger. Freud's "Das Unheimliche" (1919) derived the uncanny from the return of the familiar, *heimlich*, in estranged form, and found among its sources the double, the compulsion to repeat, and the confusion of animate and inanimate, illustrated by E. T. A. Hoffmann's "Der Sandmann" (1816), whose hero loves the automaton Olimpia. The bartender's encounter with the young man is uncanny in the strictest Freudian sense: the most familiar of all beings, oneself, returned as a stranger. Hoffmann's automaton anticipates the zombies of the title, persons who may be mechanisms. The narrator inhabits, at the end, a world of Olimpias.

§ 7.5

Repetition Compulsion and the Death Drive

In *Jenseits des Lustprinzips* (1920), Freud described the infant's *fort-da* game—casting away and retrieving a spool—as the conversion of a passively suffered experience, the mother's departure, into an actively staged one. He generalized the observation into the repetition compulsion and posited a death drive tending to restore an earlier state of things.

The narrator's career is the *fort-da* game played with time. Every suffering inflicted on Jane—seduction, abandonment, the loss of the child—is re-enacted by the bartender as his own action. The seduced becomes the seducer; the bereaved becomes the abductor. Passive trauma is converted into active mastery by being staged from the other side. And the loop is the death drive in physical form, a return to an earlier state that does not merely tend toward restoration but achieves it.

§ 7.6

Lacan: Mirror, Name-of-the-Father, Foreclosure, Sinthome

Lacan's mirror stage (1936; 1949) locates the formation of the ego in the infant's jubilant identification with its specular image, an alienating identification with an image of itself as whole. The bartender recruits his younger self by presenting it with an image of its own future completeness, himself. The mirror stage is staged across time.

In Seminar III (1955–56) and in "D'une question préliminaire à tout traitement possible de la psychose" (drafted 1957–58), Lacan held that the subject enters the symbolic order through the paternal metaphor, the substitution of the Name-of-the-Father for the desire of the mother. Where that signifier is foreclosed (*Verwerfung*), psychosis results, and what is foreclosed from the symbolic returns in the real. Jane's genealogy forecloses any father other than herself. The signifier of paternity is not repressed but absent. And what is foreclosed returns in the real exactly as Lacan predicted: the father appears not as a symbolic function but as a real stranger who seduces her, and who is the real of herself.

The Temporal Bureau supplies what Lacan called a *suppléance*, a supplementary structure that stabilizes a psychotic subject: an impersonal law of regulations, missions, and timestamps. Seminar XXIII (1975–76) theorized the *sinthome* as a fourth ring that knots together the three registers of Real, Symbolic, and Imaginary when the paternal knot has failed, and found its paradigm in Joyce's writing. The ouroboros ring is a literal sinthome, the fourth ring that holds the narrator together. The Unmarried Mother's confessional writing is a further sinthomatic practice, a Joycean binding through text.

§ 7.7

The Inverted Schreber

Daniel Paul Schreber's *Denkwürdigkeiten eines Nervenkranken* (1903), the memoir of a Saxon appellate judge, records a delusional cosmology of extraordinary coherence. Schreber believed that he was being subjected to *Entmannung*, unmanning, a transformation into a woman, so that he might be impregnated by divine rays and give birth to a new race. After a world catastrophe, the people around him were no longer real human beings but *flüchtig hingemachte Männer*, "fleeting-improvised men," hastily produced by miracle and lacking genuine existence. Freud's 1911 study read the case through repressed homosexual wish and projection. Lacan read it through foreclosure of the Name-of-the-Father and a structural push toward the woman.

The homology with Heinlein's story is structural and fourfold.

1. **Sex transformation as a condition of generativity.** Schreber must become a woman to bear the new humanity; Jane must become a man to beget herself.
2. **Relation to a progenitive power.** Schreber is impregnated by God; Jane by her own future self, who occupies the place of the god of her life.
3. **Derealization of others.** Schreber's fleeting-improvised men are Heinlein's zombies, people who appear to be persons but lack genuine being.
4. **A world after catastrophe.** Schreber's *Weltuntergang* corresponds to the narrator's emptied world at the end, in which there is no one present but himself.

The inversion is equally exact. Schreber moves from male to female, Jane from female to male. Schreber's progeny is plural, a new race; Jane's is singular, herself. The story is a paranoid cosmology transposed into hard science fiction, in which the delusion is not a delusion. This is a recurrent Heinleinian procedure. In "They" (1941) the paranoid's world is confirmed. In "—All You Zombies—" the psychotic's world is constructed and then shown, with impeccable causal logic, to be the only world consistent with the narrator's origin.

A history belongs inside this homology. Schreber's father, the Leipzig physician Moritz Schreber, was a widely read authority on child-rearing who prescribed orthopedic devices to hold children's bodies upright, together

with regimes of strict physical and emotional discipline. William Niederland, in studies begun in 1959, and Morton Schatzman, in *Soul Murder* (1973), argued that he applied these methods to his own children and that the son's persecutory "miracles" transcribe them; Schatzman traced to the father the "soul murder" that the *Memoirs* attribute to Schreber's psychiatrist, Paul Flechsig. Han Israëls (1989) contested how far the evidence supports that reading. On either account, the real father in Schreber's history was overwhelmingly present while, in Lacan's terms, the symbolic father was foreclosed. The inversion with Jane is thereby completed. Schreber's real father was an outside authority of crushing weight; Jane's real father is herself. In both cases, what cannot be symbolized as paternity returns in the real, as persecutor in the one and as seducer in the other.

§ 7.8

Anti-Oedipus: The Collapsed Triangle

Gilles Deleuze and Félix Guattari's *L'Anti-Œdipe* (1972) opens with Schreber and indicts psychoanalysis for reducing desire to the triangle of "mommy-daddy-me." Against Oedipus it sets an orphan unconscious, self-produced, exemplified by Antonin Artaud's late declaration that he is at once his own son, father, mother, and himself, a refusal of the idiotic circuit of parental generation.

Heinlein's narrator collapses the Oedipal triangle to a point. This degenerate triangle is neither Oedipal—there is no triangle—nor schizo-revolutionary—there are no flows, only a closed circuit. The loop is an Oedipus of one: absolute familialism in a family with a single member. The story thus confirms Deleuze and Guattari's diagnosis that the family is a closed loop, and refutes their utopia, since self-production, once absolutized, proves not to be liberation but enclosure. The orphan unconscious, realized, is not a desiring-machine but an ouroboros.



THE PERSON IN THE LOOP

Identity, Self-Reference, and Non-Well-Founded Genealogy

The narrator satisfies every standard criterion of personal identity across his stages, and that is precisely his predicament. Identity holds too well. This chapter shows that the loop is formally a non-well-founded structure, a token that reproduces itself as a token, and that the theories of personal identity most hospitable to it are also the ones that best diagnose its pathology.

§ 8.1

Locke, Reid, and the Transitivity of the Loop

Locke's chapter "Of Identity and Diversity," added to the second edition of the *Essay* (1694, II.xxvii), defines the person as a forensic term and locates personal identity in consciousness extended backward by memory. Thomas Reid's objection in the *Essays on the Intellectual Powers of Man* (1785, III.6), the case of the brave officer, showed that memory is not transitive: the general remembers the young officer, who remembered the flogged boy, but the general does not remember the boy. Joseph Butler (1736) charged the memory criterion with circularity, since genuine memory presupposes the identity it is meant to constitute.

In the story the memory chain is transitive throughout: the narrator remembers being the bartender, the young man, Jane, and, through the orphanage records, the infant. Reid's objection is answered not by a theory but by a fact of the narrative. But the transitive chain, closed into a loop, yields something Reid never considered: co-presence. Two stages of the same person occupy the same room at the same external time—the young man and the bartender in 1970, the seducer and Jane in 1963. Memory connects them in one direction only. The bartender remembers being the young man; the young man does not yet remember being the bartender. Within a single scene, consciousness extends asymmetrically across a single person.

Butler's circularity is likewise realized. Sydney Shoemaker's notion of quasi-memory (1970)—memory-like states that do not presuppose identity—was introduced to escape it. The bartender possesses something stranger: a memory of being recruited that guides him in recruiting. His memory functions as a script. What he remembers having had done to him, he does. Memory here is not a record of the past but an instruction for the present.

§ 8.2

Four-Dimensionalism: Worms, Stages, and Self-Interaction

Perdurantism holds that persons are four-dimensional wholes composed of temporal parts, as a road is composed of stretches. Quine's "Identity, Ostension, and Hypostasis" (1950) gave the view its canonical modern statement; Theodore Sider's *Four-Dimensionalism* (2001) distinguished the worm theory, on which a person is the whole spacetime worm, from the stage theory, on which a person is an instantaneous stage related to other stages by temporal counterpart relations.

On the worm theory, the narrator is a single four-dimensional object whose temporal parts interact causally with one another. The seduction is one temporal part impregnating another. This is a phenomenon without parallel in ordinary perdurance: reproduction as an internal relation among the parts of a single object. Call it **auto-mereological reproduction**. On the stage theory, the narrator is a series of stages linked by counterpart relations; the story then depicts stages that are each other's counterparts and each other's parents. Either way, Heinlein's 1939 worm of "Life-Line" has acquired a property no worm possesses: it generates its own beginning.

§ 8.3

Parfit's Glass Tunnel and Its Closure

Derek Parfit's *Reasons and Persons* (1984) argued that personal identity is not what matters in survival. What matters is psychological connectedness and continuity, which can hold in degrees and can, in cases of fission, hold between one person and two successors. In a celebrated passage, Parfit described how accepting this view changed his relation to his own life: he had felt himself imprisoned in a glass tunnel, moving faster each year toward its end; after his conversion, the walls of the tunnel disappeared, and he lived in the open air, less concerned with his own death and more concerned with the lives of others.

The narrator's case is the exact inversion of Parfit's liberation. Parfit's reductionism loosens identity and dissolves the walls of the tunnel into the open air of others. The narrator's identity is maximally tight—identical across every role, connected by memory in every direction available—and his tunnel is not dissolved but closed into a ring. The convergence with Korzybski noted in Chapter 2 is now exact. Both counsel the same therapy: treat the dated selves as related but distinct, and attend to the relations rather than to

the identity. Parfit and Korzybski would advise the narrator to regard Jane as someone he loved rather than someone he is. The story's tragedy is that the narrator cannot follow this advice, because the loop makes identity not a doctrine but a structure.

§ 8.4

The Genealogical Quine Atom

Standard set theory forbids self-membership. The Axiom of Foundation in Zermelo–Fraenkel set theory requires that every nonempty set contain an element disjoint from it, which excludes any set x such that $x \in x$, and any infinite descending membership chain. In New Foundations and in non-well-founded set theories, by contrast, there can be *Quine atoms*: sets identical with their own singletons. Marco Forti and Furio Honsell (1983) and Peter Aczel's *Non-Well-Founded Sets* (1988), with its Anti-Foundation Axiom, provided a rigorous theory of such structures; Jon Barwise and John Etchemendy's *The Liar* (1987) and Barwise and Lawrence Moss's *Vicious Circles* (1996) applied it to circular phenomena from semantic paradox to self-referential states.

Genealogy, normally, is well-founded: a family tree is a directed acyclic graph, and every lineage traced backward terminates or recedes without return. Jane's genealogy violates foundation. Let $P(x)$ denote the set of persons who are x's parents. For Jane:

$$P(j) = \{j\}$$

Jane is a **genealogical Quine atom**: the person whose parent-set is her own singleton. The formal analogy is not ornamental. Aczel's framework represents such structures as graphs with cycles and proves that each such graph has a unique decoration. The uniqueness of Jane's solution is therefore not a narrative accident but a formal property of the structure: the loop determines its occupant.

Genealogists use the term *implex* (German *Ahnenschwund*, "ancestor loss") for the reduction in the number of distinct ancestors caused by consanguineous marriage. A person has, in principle, two parents, four grandparents, and 2^n ancestors at the nth generation; in an inbred lineage the count collapses. Jane represents total implex: 2^n ancestors at every gener-

ation reduced to one. The coefficient of inbreeding of unity derived in Chapter 6 is the genetic expression of this genealogical fact.

§ 8.5

Strange Loops, Quines, and the Collapse of Type and Token

Douglas Hofstadter's *Gödel, Escher, Bach* (1979) and *I Am a Strange Loop* (2007) argued that the self is a "strange loop," a level-crossing pattern of self-reference analogous to the Gödel sentence that asserts its own unprovability, or to M. C. Escher's lithograph *Drawing Hands* (1948), in which two hands draw each other into existence. Jane is Hofstadter's strange loop literalized as causation. But Hofstadter insisted that the strange loop of the self is social: each self contains coarse-grained models of others, and persons partially live in one another. Jane's loop contains no one but herself. She is the degenerate case of a strange loop, one that has achieved self-reference by abolishing reference to anything else.

In computation, a *quine* is a program whose output is its own source code. Stephen Kleene's recursion theorem (1938) guarantees that such self-reproducing programs exist for any sufficiently expressive language. John von Neumann's *Theory of Self-Reproducing Automata* (edited posthumously by Arthur Burks, 1966) showed how a machine could construct a copy of itself by means of a description that is both executed and copied. Charles Sanders Peirce's distinction between *type* and *token* (1906) clarifies what is distinctive about Jane. Von Neumann's automaton and the computational quine reproduce their *type*: the output is a new token of the same pattern. Jane reproduces her *token*: the output is not a copy of her but her. She is a **token-reflexive replicator**, the only conceivable reproducer whose offspring is numerically identical to the parent.

§ 8.6

Kant's Natural End and the Limit of Autopoiesis

Kant's *Kritik der Urteilskraft* (1790, §64) defined a natural end, an organism, as a thing that is cause and effect of itself, *obgleich in zwiefachem Sinne*, though in a twofold sense. His example was the tree. A tree generates another tree of its species, and so produces itself *as a species*; it grows, and so produces itself *as an individual*; and its parts produce one another reciprocally. Humberto Maturana and Francisco Varela's concept of *autopoiesis* (1972; English 1980) defined the living cell as a network that continuously produces the components that produce it. Niklas Luhmann's *Soziale Systeme* (1984) extended autopoiesis to operationally closed social systems.

Kant's qualification, "in a twofold sense," is the key to the story. Self-production as a species—type-level self-generation—is the ordinary condition of living things. Self-maintenance as an individual—token-level self-production through growth—is likewise ordinary. Jane collapses the twofold sense into one. She produces herself as an individual by the process through which organisms produce themselves as a species. The collapse of Kant's distinction is the exact biological correlate of the Fallacy of Autogenetic Division developed in Chapter 12: a property legitimately predicated of the kind is transferred to the token, and the token becomes uninhabitable.

The Temporal Bureau is, in Luhmann's terms, an operationally closed social system that reproduces its membership through its own operations, here through recruitment loops in which agents enlist their own younger selves. It is autopoiesis raised to the level of an institution and closed in time, a bureaucracy of monads.



FATE WITHOUT AMOR FATI

Freedom, Predestination, and Recurrence

The narrator lives Nietzsche's eternal recurrence without Nietzsche's affirmation, and the story thereby isolates the condition Nietzsche called the most extreme form of nihilism. Yet the narrator remains, on the best compatibilist analysis, free and responsible, and the burden of that responsibility, owed to no one but himself, is the precise texture of his despair.

§ 9.1

Ancient Fatalism: The Sea Battle, the Master Argument, the Cylinder

Aristotle's *De Interpretatione* 9 asked whether a statement about a future contingent—that there will be a sea battle tomorrow—is now true or false, and whether its truth would make the battle necessary. Diodorus Cronus's Master Argument, reported by Epictetus (*Discourses* II.19), derived from the necessity of the past that only what is or will be actual is possible. The story answers both: every proposition about the narrator's future is already true, since he remembers it, and yet nothing compels his acts except his own desires.

Chrysippus's cylinder, preserved in Cicero's *De fato* (42–43) and Aulus Gellius (*Noctes Atticae* VII.2), offers the best ancient model of the narrator's agency. A cylinder pushed down a slope requires an external impulse to start rolling, but rolls according to its own form. Fate supplies the push; the agent's character determines the motion. In the story the loop supplies the push, and Jane's character—her resentment, her loneliness, her need for revenge and then for vocation—supplies the motion. The Stoics also taught, by Nemesius's report (*De natura hominis* 38), that after each cosmic conflagration the same Socrates would again marry the same Xanthippe and again be accused by the same Anytus. Eudemus, as quoted by Simplicius, ascribed to the Pythagoreans the claim that he would again lecture to his students holding the same staff. The bartender serving the same young man the same drink is the Pythagorean recurrence localized in a single bar.

§ 9.2

Augustine's Refutation of the Circle and Boethius's Eternity

In *De civitate Dei* XII.14, Augustine rejected the philosophers' cyclical "circuits" of time as *falsi circuli*, false circles, on the ground that Christ died once for all and will not die again. In the same book (XII.21) he wrote the sentence that Hannah Arendt would make the motto of her political philosophy: *Initium ergo ut esset, creatus est homo, ante quem nullus fuit*—that there might be a beginning, man was created, before whom there was no one. The story realizes, on the scale of a single person, the pagan circuit Augustine refuted, and with it loses the *initium*. Jane is the being before whom there was no one but herself, and therefore no beginning.

Boethius's *Consolatio Philosophiae* (V.6) defined eternity as *interminabilis vitae tota simul et perfecta possessio*, the complete, simultaneous, and perfect possession of unending life, and reconciled divine foreknowledge with human freedom by placing God's knowledge in an eternal present from which all times are seen at once. Knowledge does not cause what it knows. The bartender's foreknowledge is Boethian in structure—he knows because he has seen—but its modality is circular rather than simultaneous. The narrator possesses his life not *tota simul* but *tota recurrens*, as a whole that returns. And unlike Boethius's God, he is not outside the life he knows. His foreknowledge is memory.

§ 9.3

Predestination: Calvin, Slusser, and the Elect

Calvin's *Institutes* (III.21) set out the doctrine of double predestination: God elects some to salvation and reprobates others, by an eternal decree. Luther's *De servo arbitrio* (1525), answering Erasmus's *De libero arbitrio* (1524), denied that the will is free in matters of salvation. George Slusser's reading of Heinlein's oeuvre through a secularized Calvinism (1976, 1977) found in it a recurrent division between an elect—competent, self-reliant, knowing—and a reprobate mass.

"—All You Zombies—" is Slusser's thesis in its purest form. The narrator is elect in the strict sense: chosen, recruited, and fixed in his destiny by a decree that he himself issues from his own future. The zombies are the reprobate, the mass outside the circle of election, whose lives the narrator cannot comprehend. The 2014 film's title names the doctrine directly. What Slusser's reading does not anticipate is the inversion: the elect here is elect only to solitude, and his election is indistinguishable from the reprobation he projects upon others.

§ 9.4

Nietzsche's Recurrence and the Absence of Amor Fati

In *Die fröhliche Wissenschaft* §341, Nietzsche imagines a demon who tells you, in your loneliest loneliness, that you will live this life again innumerable times, every pain and joy in the same order. Would you curse the demon, or call his announcement divine? *Also sprach Zarathustra* ("Vom Gesicht und Rätsel") stages the doctrine at a gateway called the Moment, where two eternal paths meet; the dwarf, the spirit of gravity, glibly observes that time itself is a circle, and Zarathustra rebukes him for making it too easy. The affirmative response is *amor fati*: to want nothing to be otherwise, forward, backward, or in all eternity (*Ecce Homo*; *Fröhliche Wissenschaft* §276). In notes posthumously collected as *Der Wille zur Macht* (§55), Nietzsche described existence without meaning or aim, yet recurring inevitably without finale into nothingness, as the most extreme form of nihilism.

The narrator is in the position of Nietzsche's demon's interlocutor, with one difference: the demon is himself. His response is neither a curse nor an affirmation but melancholy. He does not will the loop; he endures it, and he longs for what it took from him. He is thus closer to the dwarf than to Zarathustra: he knows that time is a circle, and the knowledge weighs nothing. The story isolates recurrence without *amor fati*, which is to say, Nietzsche's most extreme nihilism, lived from inside. The subtitle of *Ecce Homo*, *Wie man wird, was man ist*, "how one becomes what one is," descends from Pindar's *genoi' hoios essi mathōn* (*Pythian* 2.72), "become such as you are, having learned what that is." Jane fulfills the Pindaric precept with horrifying literalness. She becomes what she is, having learned it by meeting it.

Albert Camus's *Le Mythe de Sisyphe* (1942) concluded that one must imagine Sisyphus happy, his eternal labor redeemed by lucid revolt. The narrator is Sisyphus without revolt and without the rock's indifference to him, since the rock is himself. Camus's absurd hero confronts a silent universe. Heinlein's narrator confronts an echoing one.

§ 9.5

Compatibilism and the Loop as Counterfactual Intervener

Harry Frankfurt's "Alternate Possibilities and Moral Responsibility" (1969) argued that an agent may be morally responsible for an act even if he could not have done otherwise, provided he acts on his own desires. Frankfurt's cases feature a counterfactual intervener poised to ensure the act should the agent waver, who never in fact intervenes. The loop functions as such an intervener. Jane acts throughout on her own desires; were she to waver, the consistency of the history would preclude the wavering; but she never wavers. On Frankfurt's analysis, she is fully responsible.

Lewis (1976) analyzed the time traveler's ability to kill his grandfather as relative to facts: he "can" relative to his skills and opportunity, and "cannot" relative to the fact that the grandfather survived. Kadri Vihvelin (1996) contested the compatibility, and Sider (2002) refined the counterfactual analysis. The story occupies the space their debate defines. The seducer "can" refrain, in the sense of having the capacities to do so, and "cannot," in the sense that he remembers having been seduced. Both senses are true together.

What the compatibilist analysis obscures is the peculiar shape of the narrator's responsibility. He is responsible for his own seduction, his own abandonment, the abduction of his own child, and his own recruitment. Every wrong done to him he has done. Byung-Chul Han's *Müdigkeitsgesellschaft* (2010) described the late-modern achievement-subject who exploits himself and calls it freedom, master and slave in a single person. The narrator is that subject avant la lettre: agent and victim of his own exploitation, owing accountability to no one but himself, and therefore unable to forgive or be forgiven by anyone.

The seduction shows what the identity posture conceals. Indexed in Korzybski's manner (Ch. 2.1), the encounter of April 1963 is between a man of twenty-five, who knows the girl's history, loneliness, and future completely, and a seventeen-year-old orphan who knows nothing of him; every asymmetry of age, knowledge, and power runs one way, and an agent of the Bureau arranges the meeting to secure a conception. On Parfit's account (Ch. 8.3), the psychological connections between such distant stages are weak enough that the man's treatment of the girl approaches, in moral weight, his treatment of another person. The narrator's insistence that he is all of them therefore does specific work: it converts an adult's seduction of a minor, arranged for an institutional purpose, into a self-relation, and so removes the

victim from view. The copula of identity launders the wrong. Arendt's observation that no one can forgive himself finds its sharpest application here, for the only person who could forgive the man is the girl, and the loop has made her him.

§ 9.6

Kierkegaard's Self and Heidegger's Thrownness

Kierkegaard's *Sygdommen til Døden* (1849) defines the self as a relation that relates itself to itself and, in so doing, relates itself to another, the power that established it. Such a self, he reasons, must have been established either by itself or by another. He then argues from the phenomenology of despair. If the self had established itself, there could be only one form of despair: not willing to be oneself. The existence of the second form—in despair willing to be oneself—shows that the self was established by another, since it expresses the self's inability to rest in itself.

The story is a Kierkegaardian experiment. Its narrator is the self established by itself. On Kierkegaard's logic, such a self should be capable only of the first form of despair, the despair of weakness that does not will to be oneself. And this is exactly what the coda shows: the narrator longs to be other than he is, to have back the girl he was. His despair is not defiance but grief. The story confirms Kierkegaard's taxonomy by producing the one subject to which his counterfactual applies.

Heidegger's *Sein und Zeit* (1927) defines Dasein by thrownness (*Geworfenheit*): it exists as already delivered over to a situation whose ground it did not lay. Existential guilt (§58) is being-the-basis of a nullity, the condition of a being that can never get back behind its thrownness. Jane is the one Dasein that gets back behind its thrownness. She carries herself, as an infant, into 1945 and leaves herself on a doorstep. Her thrownness is her own projection (*Entwurf*); her past is her future's act. What she finds behind her thrownness is herself, and the result is not freedom from guilt but its absolutization. The narrator is the basis not of a nullity but of his entire existence, and therefore guilty of all of it.

Heidegger's later *Brief über den Humanismus* (1947) writes existence as *Ek-sistenz*, a standing-out into the openness of being. Richard of St. Victor, as Chapter 12 will show, had already analyzed *existentia* as *ex-sistere*, being-from. Jane does not ek-sist. She in-sists: her being stands not out toward the other but back into itself.



THE SUBJECT AND ITS ZOMBIES

Solipsism, Other Minds, and the Apostrophic Inversion

On every sense of “zombie” current in 1958 or available since, the term applies more exactly to the narrator than to the reader he addresses. The closing epithet is a self-description displaced outward, and the displacement is the mechanism of solipsism itself.

§ 10.1

Descartes at the Window

In the Second Meditation, having established the certainty of his own existence, Descartes looks out of the window at men passing in the street and asks what he actually sees beyond hats and coats, *sub quibus latere possent automata*, under which automata might be hidden. He judges that they are men, but the judgment is an inference, not a perception. The *cogito* secures one existence and leaves every other in question. This is the founding scene of the problem of other minds, and Heinlein's coda restages it. The narrator, having secured with total certainty the provenance of his own existence, looks out at everyone else and sees only the hats and coats of zombies.

The story thus produces a genealogical *cogito*. Descartes's first certainty is that he thinks; the narrator's is that he begot himself. In both cases the first certainty isolates. The Cartesian subject must prove God in order to recover the world; the narrator has no such recourse, since the only god available to him is himself.

§ 10.2

From Analogy to Grammar: Mill and Wittgenstein

John Stuart Mill's *An Examination of Sir William Hamilton's Philosophy* (1865) proposed the argument from analogy: I infer other minds because others have bodies like mine and behave as I do when I feel. The argument's weakness, generalizing from a single case, is magnified to absurdity in the narrator's situation. His single case is the only case he knows from the inside, and it is a case unlike any other. Every other person has parents who are not themselves. The analogy therefore fails at the root: the narrator cannot infer others' inner lives from his own because his own is constituted by a relation no one else bears.

Wittgenstein's *Tractatus* (5.6–5.641) holds that the limits of my language mean the limits of my world, that the metaphysical subject is not in the world but is its limit, and that solipsism, rigorously carried through, coincides with pure realism: the self shrinks to an extensionless point and there remains the reality co-ordinated with it. The *Philosophical Investigations* (1953, §§243–315) dismantles the private language on which solipsism depends, and §293's beetle in the box shows that a private object drops out of the language game as irrelevant. The narrator's condition illustrates the Tractarian coincidence of solipsism and realism with eerie precision. His

solipsism is not a theory but a structural fact about his origin, and as such it is perfectly realistic. At the same time, the *Investigations* show what is wrong with it. The narrator's final address to the reader is itself a move in a public language game; he cannot even formulate his solitude without presupposing the community of speakers he denies.

§ 10.3

Fichte, Schopenhauer, Stirner

Fichte's *Grundlage der gesammten Wissenschaftslehre* (1794) begins from the I that posits itself absolutely and then posits the not-I in opposition to itself. The narrator is the Fichtean I in narrative form: self-positing, and positing others only as the not-I, the zombies, whose being is derivative of his. Schopenhauer opened *Die Welt als Wille und Vorstellung* (1819) with "the world is my representation," and in §19 considered theoretical egoism, which treats all phenomena except one's own individual as phantoms. Such solipsism, he conceded, could never be refuted by proof, but as a serious conviction it could be found only in a madhouse. He compared it to a small frontier fortress, impregnable, whose garrison can never sally forth, so that one may pass it by. The narrator's loop is that fortress: impregnable, and with a garrison of one who can never leave.

Max Stirner's *Der Einzige und sein Eigentum* (1844) opens and closes with the Goethean line *Ich hab' mein' Sach' auf Nichts gestellt*, I have set my cause on nothing. The unique one owns himself and acknowledges no higher cause. The narrator is *der Einzige* in the most literal sense available: the only one, owner of his own origin, his cause set on nothing but himself. Stirner's egoist, however, enjoys and consumes the world. The narrator inherits Stirner's sovereignty without Stirner's appetite.

§ 10.4

A Genealogy of the Zombie

The *zonbi* of Haitian Vodou, its name variously derived from Kongo *nzambi* (god) or *zumbi* (fetish), is a person whose soul has been captured by a *bokor*, a sorcerer, so that the body labors without will at another's command. American culture absorbed the figure through W. B. Seabrook's *The Magic Island* (1929), Zora Neale Hurston's ethnography *Tell My Horse* (1938), and films from *White Zombie* (1932) to *I Walked with a Zombie* (1943). By the 1950s the zombie had become a figure of the will-less conformist. David Riesman's *The Lonely Crowd* (1950) described the shift from inner-directed to other-directed character; William H. Whyte's *The Organization Man* (1956) diagnosed corporate conformity; Jack Finney's *The Body Snatchers* (1954–55) and its 1956 film imagined neighbors replaced by emotionless duplicates; Heinlein's own *The Puppet Masters* (1951) imagined human beings ridden by parasitic slugs. A bar story of 1958 also invites a lighter resonance: the Zombie was, since 1934, a celebrated tiki cocktail.

That genealogy passes over the history the figure carries. The *zonbi* took shape in Saint-Domingue, where French colonists held hundreds of thousands of enslaved Africans to forced labor on sugar and coffee plantations, and scholars of Haitian religion read it as the memory of enslavement extended past death: a body that works for another, its will removed (Dayan 1995; Lauro 2015). Haitian law registered the fear. Article 246 of the Penal Code treated the use of substances that produce a deathlike lethargy as an attempt on life by poisoning, and as murder if the victim was buried. The figure entered American popular culture during the United States' military occupation of Haiti (1915–1934), when the Marine-officered gendarmerie revived the corvée and compelled Haitian peasants to build roads (Schmidt 1971). Seabrook's *The Magic Island* was written during that occupation, and its best-known chapter describes dead men laboring in the cane fields of the Haitian-American Sugar Company. The epithet an American narrator applies to everyone else in 1959 descends from an occupier's account of Haitian workers as will-less bodies.

The history gives the Apostrophic Inversion its full precision. The zombie is a person owned by another, the exact negation of the Lockean individual who has a property in his own person (Ch. 12.6). The narrator, who owns himself absolutely, reproduces the occupier's gaze that saw others as zombies, and he is himself the figure that gaze projected: a body executing a script it did not choose.

In philosophy of mind the *zombie* is a being physically and behaviorally identical to a conscious person but lacking experience. Robert Kirk introduced the term in "Zombies v. Materialists" (1974); David Chalmers's *The Conscious Mind* (1996) made it central to the case against physicalism. Precursors include Descartes's automata and Leibniz's mill (*Monadology* §17), the thought experiment of a thinking machine enlarged so that one might walk inside and find only parts pushing one another. Heinlein's title anticipates Kirk's term by fifteen years and gives it a distinctive sense. His zombies are not beings without experience but beings without a self-relation the narrator can recognize as personhood.

§ 10.5

The Apostrophic Inversion

The thesis of this chapter, and a pillar of the dissertation, is that every one of these senses applies more exactly to the narrator than to the reader.

1. **The Haitian sense.** A *zonbi* is a body executing another's will. The narrator executes the loop's necessity and the Bureau's orders; he cannot do otherwise than he remembers having done. If the *bokor* is the loop, the Bureau, or the author, the narrator is the one who walks without will.
2. **The Cold War sense.** The conformist zombie lives by a script not his own. The narrator lives by the most rigid script imaginable, his own memory, which dictates every line he speaks to his younger self.
3. **The philosophical sense, reconstructed.** If a zombie is a being that lacks the inner relation constitutive of personhood, and if that relation is, as Chapter 12 argues, a relation to others—abaliety, recognition, natality—then the narrator, who has no others, is the zombie in the precise sense the argument requires.
4. **The Eliotic sense.** The undead crowd of the "Unreal City" are those whom death has undone, Dante's souls in the vestibule of Hell, who lived without commitment. The narrator, whose every act was committed before he performed it, is the one who never lived a life that was not already undone.

The epithet is thus a projection in the psychoanalytic sense: an attribute of the self, intolerable to it, expelled and located in the other. The mechanism by which the narrator calls others zombies is the mechanism by which solipsism sustains itself. A self that cannot recognize others as persons must

explain their apparent personhood away; it does so by attributing to them the deficit it suffers.

§ 10.6

The Question Turned Around

The closing question, an inquiry into where everyone else came from, is not rhetorical. It is a genuine question that the narrator cannot answer, and its unanswerability is the point. For a being *a se*, being *ab alio* is unintelligible. The narrator knows where he came from: from himself. What he cannot understand is how any being could come from another, since he has no experience of it.

The reader, however, can answer the question at once. Each of us came from others: from two parents who were not ourselves, from a lineage that recedes without return, from a language and a community that preceded us and address us. That answer is not a deficit but a definition. The question that the narrator poses as an indictment is, read from the other side, the reader's certificate of personhood. Twain's Satan told Theodor that nothing existed but him, a homeless thought among empty eternities. Heinlein's narrator says the same to the reader, and the reader can reply with the one fact the narrator lacks: we came from one another, and so we are not alone.

§ 10.7

The Solipsistic Turn of 1958: Bester and Heinlein

Alfred Bester's "The Men Who Murdered Mohammed" appeared in *The Magazine of Fantasy and Science Fiction* in October 1958, five months before Heinlein's story appeared in the same magazine. Bester's Henry Hassel, discovering his wife's infidelity, travels into the past to kill his grandfather and erase her, then assassinates figure after figure of history, and finds that nothing in the present changes. Each time traveler alters only his own past and, in doing so, detaches himself from the shared world, becoming a ghost who can neither be seen nor heard. Time, in Bester's formulation, is a personal matter.

The two stories, published in the same venue within half a year, constitute a solipsistic turn in late-1950s time-travel fiction. Bester's travelers become solipsists by trying to change the past; Heinlein's narrator becomes one by completing it. Bester makes mutability isolating; Heinlein makes consistency isolating. Between them, they exhaust the possibilities: on either model of

temporal causation, the subject who masters time loses the world of others. The convergence suggests that the genre, at the height of the Cold War and the organization man, had discovered in the time machine a figure for the American dream of self-sufficiency, and had found that dream uninhabitable.



COMPARATIVE CONSTELLATIONS IN SPECULATIVE AND PHILOSOPHICAL FICTION

Across seven decades of speculative and philosophical fiction, every text that confronts the self-encountering or self-originating subject divides along a single axis: whether the multiplication of the self opens onto others or closes upon itself. Heinlein's story is the pole of closure; the comparative field shows how the same materials yield plurality.

§ 11.1

Institutional Ouroboroi: Asimov, Anderson, Leiber, Lem

Isaac Asimov's *The End of Eternity* (1955) imagines Eternity, an organization outside time whose technicians adjust Reality to minimize human suffering. Its existence depends on a loop: the Eternals send Brinsley Sheridan Cooper into the primitive past, where, as Vikkor Mallansohn, he invents the temporal field that makes Eternity possible. Eternity bootstraps itself. The novel's climax refuses the loop. Andrew Harlan and Noÿs Lambent, an agent of the far future's Hidden Centuries, break the circuit, ending Eternity so that humanity, no longer protected from risk, can reach the stars. Asimov's resolution chooses an open future of many over the closed security of the institution. Heinlein's Temporal Bureau, which recruits its agents from their own earlier selves, is Eternity without Noÿs.

Poul Anderson's Time Patrol stories (from 1955; collected as *Guardians of Time*, 1960) present a police of history founded by the Danellians, far-future posthumans who establish the Patrol to secure the history that produces them. The institution protects its own origin, an institutional bootstrap. Fritz Leiber's *The Big Time* (1958) sets its action in a recuperation station outside time, staffed by "Entertainers" who comfort soldiers of the Change War between the Spiders and the Snakes. The time-soldiers regard the unrecruited populace of the "little time" with a condescension that anticipates Heinlein's epithet; the Snakes supply the ouroboros its faction; and the Entertainers parallel the hospitality and entertainment of W.E.N.C.H.E.S.

Stanisław Lem's *Dzienniki gwiazdowe* (*The Star Diaries*, from 1957) contains the comic counterpart. In the Seventh Voyage, Ijon Tichy's ship passes through gravitational vortices that duplicate him across the days of the week, so that Monday's Tichy argues with Thursday's about who should repair the rudder. In the Twentieth Voyage (1971), Tichy is drafted to direct a vast program for improving history and discovers that the program's errors and its staff of temporal agents have produced the very history it set out to correct. Lem treats the institutional loop as farce; Heinlein, as tragedy. Both see that an institution recruiting itself across time becomes the author of the disorder it polices.

§ 11.2

The Self Encounters Itself: Heinlein, Gerrold, Borges, Russ

Heinlein's own "By His Bootstraps" (1941), discussed in Chapter 2, is the information loop; "—All You Zombies—" is the genealogical loop. The progression from a notebook without author to a person without parents measures the radicalization of the bootstrap across seventeen years.

David Gerrold's *The Man Who Folded Himself* (1973) is the most direct descendant. Daniel Eakins, given a timebelt, multiplies himself across time, holds a party attended only by versions of himself, enters a sexual relationship with a male double, meets a female version, Diane, and fathers the child who will become himself. Gerrold renders explicit the erotics of self-love that Heinlein sublimates into melancholy, and his novel has been read, with justice, as a queer text in which self-multiplication becomes an exploration of desire. Yet Gerrold's ending, in which the aged protagonist attends to his own infancy, arrives at Heinlein's conclusion: a life composed entirely of oneself is a life without companions.

Jorge Luis Borges supplied the metaphysical register. In "Las ruinas circulares" (1940), a man dreams another man into existence, then discovers, as fire fails to burn him, that he too is being dreamt: the generator is generated. In "Nueva refutación del tiempo" (1944–47), Borges applied Berkeley's and Hume's idealism to time itself before conceding, in the essay's famous close, that time is the substance he is made of. In "El otro" (1972), the aged Borges meets his younger self on a bench by the Charles River, and the two cannot fully believe each other. Borges's self-encounters produce estrangement, not identity; his younger self remains other. Here lies the decisive difference from Heinlein: Borges preserves the alterity of the self across time that Heinlein's narrator cannot.

Joanna Russ's *The Female Man* (1975) follows four women—Joanna, Jeannine, Janet, and Jael—who are versions of one woman across alternative worlds. Jael, an assassin from a world at war between the sexes, gathers the others in order to recruit them to her cause. The structure is Heinlein's: one person's multiple versions meet, and the most militarized recruits the others. The outcome is opposite. Russ's variants remain distinct persons who argue, disagree, and decline Jael's offer; their multiplicity grounds a feminist solidarity rather than a solipsistic identity. Russ, who wrote searchingly about Heinlein's women, thus produced the strongest feminist rejoinder to his story's architecture: the multiplication of the self as the foundation of plurality.

§ 11.3

Androgyny Communal and Solitary: Sturgeon, Le Guin

Theodore Sturgeon's *Venus Plus* X (1960), published a year after Heinlein's story, transports Charlie Johns to Ledom, a society of persons who each bear both sets of reproductive organs and can both conceive and impregnate. The revelation that the Ledom are surgically produced rather than naturally evolved parallels Jane's surgical reconstitution. But Sturgeon's hermaphrodites live in a dense communal order, raising children collectively; their androgyny distributes reproduction across a society rather than concentrating it in a self.

Ursula K. Le Guin's *The Left Hand of Darkness* (1969) imagines the Gethenians, ambisexual beings who enter a sexual phase, kemmer, in which they become either male or female, so that a single person may be the mother of some children and the father of others. Le Guin's envoy records, with deliberate shock, that the king was pregnant. Gethenian ambisexuality embeds persons in kinship webs of hearth and domain, binding them to others more tightly, not less. Le Guin's *The Dispossessed* (1974) adds a temporal ethics: Shevek's General Temporal Theory unifies Sequency (linear time) and Simultaneity (cyclical time), and its human corollary is the promise, the bond across time between persons. A promise is time bound to another. Jane's loop is time bound to oneself, the parody of fidelity.

§ 11.4

Fixity and Foreknowledge: Chiang, Vonnegut, Moorcock, Marker

Ted Chiang's "Story of Your Life" (1998) gives its linguist, Louise Banks, a simultaneous mode of consciousness acquired through an alien language, so that she knows, before her daughter is conceived, that the daughter will die young, and chooses the life anyway. "The Merchant and the Alchemist's Gate" (2007) presents a fixed past that can be visited but not changed, and finds consolation in knowing it better. "What's Expected of Us" (2005) imagines a device that proves free will illusory and records the resulting despair. Chiang's protagonists achieve the *amor fati* that Heinlein's narrator lacks, and the difference is precisely the other: Louise affirms her fixed future because it contains her daughter. Kurt Vonnegut's *Slaughterhouse-Five* (1969) makes eternalism a consolation through its Tralfamadorians, for whom every moment always exists; Billy Pilgrim, unstuck in time, attains a quietism that the narrator of 1958 cannot reach.

Michael Moorcock's *Behold the Man* (1966 novella; 1969 novel) sends Karl Glogauer to Palestine in 28 CE in search of the historical Jesus. Finding that Jesus is not what scripture records, Glogauer assumes the role and is crucified, becoming the founder of the faith he sought. Moorcock's religious bootstrap and Heinlein's genealogical one share a structure—the seeker becomes the origin sought—but Glogauer's self-sacrifice opens onto the multitudes of a religion. Chris Marker's *La Jetée* (1962), the photo-roman in which a man is drawn back to the airport pier where, as a child, he witnessed his own death, gives the predestination loop its most austere visual form. Marker's loop closes on death; Heinlein's closes on birth.

§ 11.5

Derived Persons: Shelley, Lem, Fitzgerald

Mary Shelley's *Frankenstein* (1818) presents a being made without a mother by a single male creator, the fantasy that Ellen Moers (1976) and Anne Mellor (1988) read as a usurpation of female generativity. Its creature asks, in the most famous series of questions in the novel, "Who was I? What was I? Whence did I come? What was my destination?" Jane's self-generation does not usurp the female but incorporates her; it is an autogenetic androgyne where Shelley's is a paternal monster. Yet the creature, who knows exactly whence he came, and the narrator, who knows exactly whence he came, end alike, outcast and alone.

Lem's *Solaris* (1961) supplies the decisive counterexample for the Abaliety Thesis. The ocean of Solaris generates "visitors" from the memories of the station's crew; Kelvin's visitor, Rheya, is materialized from his memory of his dead wife. She is wholly derived from another, the most abalietal being in literature. Yet she becomes a person: she suffers, chooses, and finally sacrifices herself. Lem shows that total derivation from another does not preclude personhood; Heinlein shows that total derivation from oneself does. Together they constitute a controlled experiment.

F. Scott Fitzgerald's *The Great Gatsby* (1925) gives the American mythology of self-making its canonical sentence: "The truth was that Jay Gatsby of West Egg, Long Island, sprang from his Platonic conception of himself." Gatsby is self-begotten in the idiom of the self-made man; Jane is self-begotten in fact. Both lineages end in the same isolation—Gatsby's empty funeral, the narrator's empty dark—and Chapter 12 traces the shared American root.

§ 11.6

Screen Ouroboroi

Cinema and television have extended the constellation. *The Terminator* (1984) sends a son's soldier back to father the son; *Back to the Future* (1985) plays the Oedipal loop as comedy; *Primer* (2004) multiplies nested loops among engineers who betray their own duplicates. The Spierigs' *Predestination* (2014), the story's adaptation, adds an agent who becomes the terrorist he hunts, literalizing the Apostrophic Inversion: the pursuer of the monstrous other is himself. *Interstellar* (2014) makes a father the ghost who guides his daughter from within a higher-dimensional structure. The German series *Dark* (2017–20) builds a whole genealogy of bootstraps, including a mother and daughter who are each other's mother, and names its structure "the knot." Christopher Nolan's *Tenet* (2020) ends by revealing that its protagonist founded the organization that recruited him, the institutional ouroboros of Heinlein's Bureau.

§ 11.7

The Axis of the Field

Arrayed together, these works sort themselves along a single axis. At one pole stand the texts of closure, in which the multiplication of the self abolishes others: Heinlein, Gerrold's ending, *Predestination*, Bester. At the other stand the texts of opening, in which the multiplication or derivation of the self grounds plurality: Asimov's shattered Eternity, Borges's unbelieving younger self, Russ's dissenting Js, Sturgeon's and Le Guin's communities, Chiang's affirmed daughter, Lem's self-sacrificing visitor. The variable that determines position is not the mechanism of time travel, nor the consistency of the loop, but whether the self that meets itself remains capable of meeting anyone else.



AT THE HEINLEIN LIMIT

The Abaliety Thesis and a Heterogenetic Theory of Personhood

The preceding chapters converge on a single result: the person is constitutively *ab alio*, and Heinlein's story demonstrates it by constructing the one subject who is not. This chapter grounds the result in a regularity governing every coherent discourse of self-origination, recovers its theological pedigree, states it formally, defends it against seven objections, and extends it to political anthropology, recursive information systems, and the formal study of kinship in time.

§ 12.1

The Law of the Qualifier

Every coherent predication of self-origination in the Western tradition carries a qualifier, and every such qualifier reintroduces an other. Plotinus, venturing that the One is in some sense its own cause, instructs his reader to attach "as it were" (*hoion*) to each such predicate (*Enneads* VI.8.13). Anselm grants that the supreme nature exists through and from itself only after denying that it is its own efficient cause (*Monologion* 6). Descartes allows that God is cause of himself only in a certain sense and, pressed by Arnauld, retreats to the analogy of formal causation. Spinoza confines the *causa sui* to the one substance, so that everything else exists in another (*Ethics* I, Ax. 1). Kant calls the organism cause and effect of itself *obgleich in zwiefachem Sinne*, though in a twofold sense, assigning self-production to the species and self-maintenance to the individual (*Kritik der Urteilskraft* §64). Marx credits humanity with a birth through itself, but credits it to the species, through the circle of the generations. Levinas writes that in paternity I do not have my child but am, in some way, my child, and warns that this "am" is not the "is" of Eleatic identity (*Le Temps et l'autre*, 1947). Hofstadter's strange loop of the self is woven from internal models of other selves. Korzybski's time-binding is accomplished by generations, never by a man.

Call this regularity the **Law of the Qualifier**: wherever self-origination is predicated coherently, the predication is qualified by a term—an analogy, a kind, a plurality, a manner of being other than identity—that relocates the origin in an other. The qualifier is the trace of abaliety inside the discourse of aseity. "—All You Zombies—" is an experiment in its deletion. It predicates self-origination of an individual, by efficient causation, in the copula of identity, without analogy, kind, or manner. The Heinlein Limit is the condition of the deleted qualifier, and the narrator's terminal solitude is what the qualifiers had been guarding against for two millennia.

The relevant passage of Marx's 1844 manuscripts, in the section on private property and communism, is the most exact philosophical anticipation of the story's structure. A being, Marx observes, regards itself as independent only when it stands on its own feet, and it stands on its own feet only when it owes its existence to itself; whoever owes another the very creation of his life lives by that other's grace. Confronting the regress of begetting—who begot my father, and his father?—Marx directs the questioner to the circular movement, sensibly perceptible within the regress, by which man repeats himself in procreation and so always remains the subject. World history, for

socialist man, is the begetting of man through human labor, and therefore the visible proof of his birth through himself. Marx's qualifier is the generic singular: the subject who remains the subject through the circle of procreation is the species. Heinlein contracts Marx's circle to a single individual. Jane is the only being who satisfies Marx's criterion of independence literally, and the story discloses that independence so achieved is indistinguishable from solitude.

Levinas's qualifier is subtler and, for the story, decisive. Paternity, on his account, gives the self a future that is at once its own and not its own: the child is myself as a stranger, and fecundity breaks the self's monotonous return to itself (*Le Temps et l'autre* IV; *Totalité et infini*, 1961, sec. IV). The formula "I am in some way my child" holds identity and alterity together only because its "am" is not the copula Korzybski indicted (Ch. 2.1). Heinlein's narrator pronounces that copula without reservation: he *is* his child in the Eleatic sense. In deleting Levinas's "in some way," the story abolishes the strangeness that made paternity, for Levinas, the paradigm of transcendence within time.

§ 12.2

Abaliety Recovered: Origin and Relation in Trinitarian Personhood

The scholastic valuation of abaliety as a creaturely deficiency is overturned by the theological tradition's own account of divine persons. Four features of Trinitarian doctrine establish that being-from-another is not a defect but a mode of personal being.

1. **Abaliety within God.** The Niceno-Constantinopolitan Creed confesses the Son as *Deum de Deo, lumen de lumine*: God from God. The Son is *ab alio* and fully divine; if derivation from another were a defect, it could not obtain within God.
2. **The unoriginated person is relational.** The Father is *a nullo*, from no one. His notional property of innascibility is negative (*ST* I, q. 33, a. 4); his personhood is constituted positively by paternity, a relation to the Son.
3. **No person is from itself.** The Father's unoriginatedness is not self-origination. Orthodoxy grants aseity to the divine nature and denies self-derivation to every person, the Father included.

4. **Persons are individuated by origin.** Richard of St. Victor analyzed *existentia* into *sistere*, what a being is, and *ex*, whence it has what it is (*De Trinitate* IV), and replaced Boethius's definition of the person as an individual substance of a rational nature with the incommunicable existence of the divine nature (IV.22). For Richard, a person's "ex" is its personal identity.

Richard also supplied the tradition's argument that personhood requires plurality. Supreme goodness entails charity, and no one is properly said to possess charity on account of a private love of himself: *oportet igitur ut amor in alterum tendat ut caritas esse possit*, love must tend toward another if it is to be charity (*De Trinitate* III.2). Supreme charity, moreover, wills that its beloved be loved by a third, the *condilectus*, with whom love is shared (III.11). Jane's love is *privatus et proprius sui ipsius amor* in the most literal sense conceivable. Her one erotic encounter is with herself; her longing in the coda is for herself; there is no third with whom her love could be shared. On Richard's analysis, she is the being for whom charity is impossible.

Aquinas supplies the logical consequence. Real relations require really distinct terms, and the divine relations are distinguished only by their opposition (*ST* I, q. 28, a. 3). The relation of a thing to itself—*idem eidem idem*—is a relation of reason only, since it requires the intellect to apprehend one thing as two (*ST* I, q. 13, a. 7). The relations among Jane's temporal stages are real causal relations, because stages are really distinct; but her relations to her father, her mother, and her child are, at the level of the person, relations of reason. Her family exists only in her representation of it. Personhood leaks out at precisely this juncture: the gap between the stage, which can be related to others, and the continuant, which can be related only to itself.

The verdict promised in Chapter 3.6 follows. Orthodoxy secured divine personhood by refusing modalism: the persons are really distinct because their relations of origin are really opposed. Jane is a modalist deity without a Trinity—one subject, many masks, no opposed relations—and so, on the tradition's own terms, a being with *personae* and no person. She occupies the position that the tradition refused to every person, human or divine: a person whose relation of origin terminates in itself.

§ 12.3

The Heterogenetic Theory of Personhood: Derivation, Address, Natality

A person is constituted by three relations to others: derivation from another, address by another, and the capacity to begin among others. The loop subtracts all three, and the narrator is what remains.

Derivation. To be a person is first to have received one's being. Marcel Mauss's *Essai sur le don* (1925) showed that the gift founds obligations to receive and reciprocate, and with them the social bond; Jacques Derrida's *Donner le temps* (1991) argued that any circle returning a gift to its giver annuls it as gift. Jane's existence is the purest such circle. Her life returns to its giver without remainder, so it was never given: she is the one being who has received nothing. Arendt defines plurality as the condition of action: humans are all the same in being human, yet in such a way that no one is ever the same as anyone else who ever lived, lives, or will live (*The Human Condition*, 1958, §1). Jane's lineage negates the condition, since every member of it is the same as she. Gilbert Simondon's doctoral thesis on individuation, defended in Paris in 1958, held that no individual is ever completely individuated: each carries a preindividual charge that opens it onto collective, transindividual life. Jane is the completely individuated individual, closed upon her origin without remainder.

Address. Fichte's *Grundlage des Naturrechts* (1796, §3) corrected the self-positing I of the 1794 *Wissenschaftslehre*, which Chapter 10.3 identified with the narrator. A finite rational being can posit itself as free only if it is summoned (*aufgefordert*) to free activity by another; the summons is what is called upbringing; the human being becomes human only among human beings. The narrator's every summons is self-issued: the bartender recruits the young man, and the confession is told to its own teller (Ch. 5.5). A summons from oneself is no summons, and the passage of Fichte's thought from 1794 to 1796 is the argument of this dissertation in miniature. Hegel holds that self-consciousness attains satisfaction only in another self-consciousness, and names spirit as the "I" that is "We" and the "We" that is "I" (*Phänomenologie des Geistes*, 1807, §§175, 177). The bartender's mastery of the young man is the degenerate limit of the struggle for recognition: victor and vanquished are identical, and recognition won from oneself is worthless because it is not other. Buber's *Ich und Du* (1923) holds that the human being becomes an I through a Thou. Bakhtin, in notes of 1961, denied that a person possesses any internal sovereign territory: to look within is to look into the eyes of another.

The narrator possesses exactly such a territory, his loop, and looking within he meets only his own eyes. Paul Ricoeur's *Soi-même comme un autre* (1990) distinguished *idem*-identity, sameness, from *ipse*-identity, selfhood, whose paradigm is keeping one's word to another. The narrator is maximally *idem* and minimally *ipse*: perfectly the same, with no one to whom his word could be kept. Robert Spaemann's *Personen* (1996) condenses the tradition into one sentence: persons exist only in the plural.

Natality. Arendt made natality—the fact that human beings enter the world by birth, as newcomers—the central category of political thought, and located in it the capacity for action, the initiation of the unforeseen (*The Human Condition* §§24, 34). Natality in her sense is not metaphysical indeterminism. A person in a four-dimensional block universe can still begin, since no one possesses memory of the future and every act is new to its agent and to others. The narrator cannot begin because his acts are already remembered; memory functions for him as a script (Ch. 8.1). Arendt further identified forgiving and promising as the faculties that redeem action from irreversibility and unpredictability, and observed that both depend on plurality: no one can forgive himself, and no one is bound by a promise made only to himself (§33). The narrator can neither forgive nor be forgiven, since his seducer, abductor, and recruiter are himself; and he can promise nothing, since his future is his past.

The three relations share a single form: each binds the self to what it is not—an origin, an addressor, a future. They are aspects of one event, the arrival of a newcomer among others. The parent from whom one derives is the first who addresses one; the first who addresses one is the first for whom one is a beginning. Call this the **triad of the newcomer**. Jane is the only human being who is never a newcomer: she has always already been there.

The narrator satisfies the psychological criteria of personhood—continuity of memory, character, and concern—while failing every relational one, which is why his case is tragic rather than merely puzzling. His one surviving desire, the longing for the girl he was, is the residue of the subtracted relations. Desire is the form that abaliety takes once its object has been abolished.

§ 12.4

Formal Statement: The Heinlein Limit and the Argument from Subtraction

Let *Par*(*y*, *x*) state that *y* is a parent of *x*, and let *Par*⁺ be the transitive closure of that relation. The ancestry of *x* is:

$$A(x) = \{\, y \mid \mathrm{Par}^{+}(y, x) \,\}$$

Genealogy is well-founded at *x* when $x \notin A(x)$. The **Heinlein Limit** is the condition in which a person's entire ancestry is that person:

$$A(x) = \{x\}$$

Sewall Wright's method of path coefficients yields a graded measure. Let $\Pi(x)$ be the set of ancestral paths that lead from *x* back to *x* without passing through *x* in between, and let $\ell(p)$ be the number of generations in path *p*. The **autogenetic fraction** of *x*, the expected proportion of *x*'s genome identical by descent from *x*, is:

$$\alpha(x) = \sum_{p \in \Pi(x)} 2^{-\ell(p)}$$

For every actual human being, $\alpha = 0$. For Jane, two paths of a single generation give $\alpha = 1/2 + 1/2 = 1$. The genetic and genealogical criteria can diverge in exotic cases: a person whose two parents both descend wholly from her has $\alpha = 1$ without satisfying $A(x) = \{x\}$, since her parents are others. The thesis turns on the genealogical criterion, because one derives from persons, not from genomes.

The argument is abductive and follows Mill's method of difference (*A System of Logic*, 1843, III.8): the story holds the psychological criteria of personhood constant and subtracts the other.

1. The narrator satisfies the Lockean criteria across all his stages: continuity of consciousness, transitive memory, stable character (Ch. 8.1).
2. He lacks derivation from another: $A(j) = \{j\}$ and $\alpha(j) = 1$ (Chs. 6.3, 8.4).

3. He lacks formative address by another: every summons he receives is his own (Ch. 5.5; §12.3).
4. He lacks natality: his acts execute his memory (Chs. 8.1, 9).
5. He exhibits the marks of depersonalization: derealization of others (Ch. 10), a psychotic structure (Ch. 7), the incapacity to forgive or promise (§12.3), and a desire whose sole object is a former self (Ch. 4.2).
6. Since (1) holds while (2)–(4) fail and (5) results, the psychological criteria are insufficient for personhood in the full sense, and the missing condition is the triad of the newcomer.
7. Personhood is therefore constitutively *ab alio*. This is the **Abaliety Thesis**.

Lem's *Solaris* supplies the converse experiment (Ch. 11.5): a being derived wholly from another, Kelvin's memory, acquires personhood through address and sacrifice. Together the two cases isolate the variable.

One further result explains the epithet. To a being whose every relation of origin is internal, persons whose origins lie elsewhere are inexplicable, since their spontaneity has no source in him. This is the Wheelerian surplus of Chapter 1.6 experienced from within. Other persons appear to such a being as bodies in motion whose motion comes from nowhere he can trace—as zombies. The Apostrophic Inversion (Ch. 10.5) is therefore structural, not merely rhetorical: it is the perceptual form that persons assume for a subject at the Heinlein Limit, and the epithet he projects outward is the name of his own condition.

§ 12.5

Objections and Replies

The Abaliety Thesis invites seven objections; each, pressed, confirms it.

The objection from consciousness. The narrator plainly thinks, feels, and suffers. Reply: the thesis distinguishes psychological personhood, which he possesses, from personhood in the full relational sense, which the Heinlein Limit subtracts; his suffering is the evidence of the subtraction, not a counterexample to it. Nor does the thesis imply that isolation diminishes the personhood of actual human beings. The hermit, the prisoner, the orphan, and the person whose capacities for communication are impaired all satisfy the triad: each was born of others, was addressed by others before being able to answer, and remains a newcomer to every other. Solitude is not the Heinlein

Limit; only genealogical closure is. The thesis grants personhood to every actual human being without exception and withholds it only from a counterfactual limit case.

The objection from sex. The thesis appears to pathologize Jane's change of sex. Reply (cf. Ch. 6.7): the loop, not the transition, causes the narrator's solitude. The change of sex is structurally the loop's only aperture onto alterity. It introduces sexual difference, the minimal other, into a worldline otherwise composed of the Same, and the narrator's longing for the girl he was is a longing for that difference. Gender self-determination presupposes a community that recognizes the self so determined; genealogical self-origination forecloses that community. The thesis affirms the former and diagnoses the latter.

The objection from autonomy. Kant makes self-legislation constitutive of rational personhood. Reply: autonomy is self-legislation, not self-origination, and its law is universal. The autonomous will legislates as a member of a kingdom of ends, a systematic union of rational beings under common laws (*Grundlegung zur Metaphysik der Sitten*, 1785). Autonomy is intrinsically plural. The narrator legislates for a kingdom of one, which is not a kingdom of ends but a solitude with a constitution. Autonomy of law is not aseity of being.

The objection from God. If God is *a se* and personal, aseity and personhood are compatible. Reply: in the Trinitarian tradition aseity belongs to the divine nature, while the persons are constituted by relations of origin among themselves (§12.2), so divine personhood confirms the thesis. For non-Trinitarian monotheisms the question of divine personhood takes other forms, and a thesis concerning finite, embodied persons need not adjudicate it.

The objection from improbability. Horwich, Lewis, and Smith establish that the loop is consistent and merely improbable (Chs. 1, 6), so nothing about personhood follows from it. Reply: the thesis does not rest on incoherence. The story functions as a limit concept, like the frictionless plane: physically unrealizable (Ch. 6.3), and for that very reason able to isolate a variable that actual cases always mix with others. Its biological impossibility marks it as fantasy; its consistency makes the fantasy instructive.

The objection from the multiverse. On the quantum treatment of David Deutsch and Michael Lockwood (Ch. 1.7), a traveler to the past arrives in a branch other than the one he left: the grandfather paradox becomes the killing of a counterpart, and the knowledge paradox dissolves because the

information carried by a loop was created by ordinary processes in some branch. Applied to the story, the seducer who reaches 1963 meets a counterpart rather than his own past; Jane's father is qualitatively like her and numerically other; her genome was produced by evolution somewhere in the multiverse. Reply: the objection concedes the thesis. It shows that the only way to make the story's events physically reasonable is to restore derivation from another. Deutsch's principle that knowledge must be created and the Abaliety Thesis are one principle applied to information and to persons; the knowledge paradox and the person paradox are a single paradox, resolved by replacing the ring with a braid. This is the Everettian therapy promised in Chapter 1.7, and Heinlein's late turn to the multiverse (Ch. 1.8) reads as his administration of it to himself.

The objection from the collective. If collectivities may originate themselves, the thesis seems to license the absorption of individuals into the collective. Reply: the Fallacy of Autogenetic Division has a converse, set out in §12.6, which the thesis condemns with equal force.

§ 12.6

The Fallacy of Autogenetic Division and Its Converse

The **Fallacy of Autogenetic Division** infers, from the premise that a collectivity produces itself through the succession of its members, that each member produces itself. Its converse, **Autogenetic Hypostasis**, infers from the same premise that the collectivity is a single self producing itself. Division deletes the qualifier by transferring self-origination from kind to token, collapsing Kant's twofold sense (Ch. 8.6); Hypostasis deletes it by treating the kind as a token. Legitimate predication preserves both the kind and the plurality of its members.

PREDICATION	BEARER OF SELF-ORIGINATION	QUALIFIER	EXEMPLARS	OUTCOME
Legitimate	A plurality across generations	Kind and plurality preserved	Kant's natural kind; Marx's species; Korzybski's time-binding; Arendt's plurality renewed by natality	Members remain *ab alio* within a self-renewing whole
Division	The individual	Deleted	The self-made man; Gatsby; Jane	Solitude at the Heinlein Limit

PREDICATION	BEARER OF SELF-ORIGINATION	QUALIFIER	EXEMPLARS	OUTCOME
Hypostasis	The collective as one self	Kind kept, plurality deleted	Hobbes's "Mortal God"; Asimov's Eternity; the Temporal Bureau	Members reduced to functionaries of a self-founding apparatus

The political anthropology of Division is what C. B. Macpherson called possessive individualism: the conception of the individual as essentially the proprietor of his own person and capacities, owing nothing to society for them (*The Political Theory of Possessive Individualism*, 1962), descended from Locke's dictum that every man has a property in his own person (*Second Treatise* §27). The narrator is the possessive individual perfected: he owns his person in the strongest sense available, having produced it. The Abaliety Thesis does refute the descriptive half of Macpherson's formula, the claim that the individual owes nothing to society for his person and capacities; it leaves open the normative question of self-ownership. Robert Nozick's *Anarchy, State, and Utopia* (1974), within the libertarian tradition with which Heinlein's later work is often associated, defends self-ownership as a thesis about rights compatible with every acknowledged form of dependence; Michael Sandel's critique of the "unencumbered self" (*Liberalism and the Limits of Justice*, 1982) presses the communitarian objection. The Fallacy of Autogenetic Division concerns neither position as such, but the inflation of a normative claim—that I own myself—into an ontological one—that I originate myself.

The American mythology of self-making is the cultural form of that inflation. Franklin's *Autobiography*, Emerson's "Self-Reliance" (1841), and Horatio Alger's *Ragged Dick* (1868) compose its canon; Fitzgerald's Gatsby, who sprang from his Platonic conception of himself, is its tragic summit (Ch. 11.5). In 1859, a century before Heinlein's story appeared, Frederick Douglass first delivered "Self-Made Men," a lecture he would revise and repeat for more than three decades. There one of the most celebrated self-made men in nineteenth-century America stated the Abaliety Thesis with precision: properly speaking, there are no self-made men, because the term implies "an individual independence of the past and present which can never exist." Douglass praised industry and self-cultivation while insisting on every person's debt to others, holding the normative and the ontological apart exactly as the present argument requires. Heinlein's narrator realizes the independence Douglass declared impossible, and confirms that it cannot be lived.

Heinlein's own competent man completes the genealogy. Lazarus Long's catalogue of the tasks a human being should be able to perform, closing with the dictum that "specialization is for insects" (*Time Enough for Love*, 1973), describes an ideal of total self-sufficiency. The narrator is its *reductio*. He has performed every office of his own life, conception and recruitment included, and the competent man who needs no one has become the man who has no one.

Hypostasis has its classical text in Hobbes, whose sovereign is an artificial person bearing the persons of all its subjects, a "Mortal God" (*Leviathan*, 1651, chs. 16–17), and its modern diagnosis in Carl Schmitt's thesis that the significant concepts of the modern theory of the state are secularized theological concepts (*Politische Theologie*, 1922). The Temporal Bureau is a Hobbesian person closed in time: a self-founding apparatus that recruits its agents from their own earlier selves and secures the history that produces it (cf. Anderson's Time Patrol, Ch. 11.1). The story shows the two errors converging. The self-made individual ends as functionary of the self-made bureau, and Division is absorbed into Hypostasis. The two figures that American social criticism of the 1950s set in opposition—the inner-directed self-made man and the other-directed organization man (Riesman 1950; Whyte 1956)—are disclosed as one ouroboros at two scales.

Anthropology supplies the institution that blocks both errors. Claude Lévi-Strauss's *Les Structures élémentaires de la parenté* (1949) interpreted the incest prohibition less as a rule forbidding marriage with mother, sister, or daughter than as a rule obliging them to be given to others, and so founded culture on exchange. Gayle Rubin's "The Traffic in Women" (1975) exposed the androcentrism of a model in which women are the gifts exchanged; the structural insight that the prohibition institutes exchange with others survives the critique. Exogamy is the institutional form of abaliety: it guarantees that every person derives from two lineages and that every union is an alliance with an other. Jane's self-union is incest at its limit—endogamy of one—and therefore the abolition of the exchange that, on Lévi-Strauss's account, constitutes society.

§ 12.7

Recursive Systems: Compilers, Models, Feeds

The Abaliety Thesis generalizes beyond persons to any self-maintaining lineage required to produce novelty: such a lineage must derive part of its input from outside itself. Three contemporary systems exhibit the Heinlein structure and its costs.

Ken Thompson's Turing Award lecture, "Reflections on Trusting Trust" (1984), described a compiler altered to plant a backdoor in the login program and to reinsert its own alteration whenever it compiled a new compiler. Once the corrupted binary existed, the alteration could be deleted from every source file and would persist indefinitely in the binary lineage—information present in every descendant and absent from every visible origin, Jane's genome in silicon. Thompson drew the moral that one cannot trust code one did not totally create oneself. The moral is autogenetic, and the demonstration refutes it, since no programmer totally creates code: every program is compiled by a toolchain inherited from others. Trust, in software as among persons, rests on derivation from others; the only alternative is a closed loop that can conceal anything.

Ilia Shumailov and colleagues (*Nature*, 2024) showed that generative models trained recursively on data produced by earlier generations of models undergo what they termed model collapse: the tails of the original distribution vanish first, and the lineage converges on a narrow, low-variance distribution that no longer represents the world from which it began. The population-genetic analogy of Chapter 6.3 is exact. Recursive self-training is selfing; the loss of the tails is the loss of heterozygosity; the terminus is a coefficient of inbreeding approaching unity. A generative lineage, like a biological one, remains viable only by continued derivation from others, in this case from data produced by human beings.

Recommender systems that generate a user's future exposure from his past selections reproduce the bartender's office: they serve the customer his own story. Eli Pariser's "filter bubble" (2011) named the hypothesized result. Its empirical extent is contested: Eytan Bakshy, Solomon Messing, and Lada Adamic (*Science*, 2015) found individual choice a stronger constraint on exposure than algorithmic ranking. Whatever its magnitude, the structure is Heinlein's, and its normative implication follows from the thesis: a system impoverishes a person in proportion as it closes the loop between what he was and what he will meet. Its epigraph is Ovid's *inopem me copia fecit*.

The general principle may be called **open derivation**: every lineage of organisms, programs, models, or persons that must generate the new must remain open to an outside it did not produce. Deutsch's principle that knowledge must be created, the evolutionary function of sex (Ch. 6.3), and the Abaliety Thesis are three statements of this single law.

§ 12.8

Chronogenealogy: A Typology of Loops

Fiction has produced enough genealogical loops to permit a typology, and the typology isolates the decisive variable: not closure as such, but closure of length one.

LOOP TYPE	EXAMPLE	GENEALOGICAL STRUCTURE	α(x)	RELATIONS TO OTHERS PRESERVED
Affinal	"I'm My Own Grandpa" (Latham and Jaffe, 1947)	Kinship by marriage forms a cycle; descent remains well-founded	0	All three
Subcritical	*Futurama*, "Roswell That Ends Well" (2001)	The subject is one of his own grandparents; other ancestry open	1/4	Partial derivation from others; address; natality limited by foreknowledge
Dyadic	*Dark* (2017–20): Charlotte and Elisabeth	Two persons, each the other's mother	1/4 each	Derivation from each other; mutual address
Institutional	Asimov's Eternity; Heinlein's Temporal Bureau; *Tenet* (2020)	An organization founds itself through recruitment	—	Members derive from others; the collective is hypostasized
Critical (Heinlein Limit)	"—All You Zombies—" (1959); *Predestination* (2014)	$A(x) = \{x\}$	1	None

The dyadic case is decisive. Charlotte and Elisabeth inhabit a closed loop, yet each derives from another and each can address the other; their tragedy is the tragedy of fate, not of solitude. Closure is therefore compatible with abaliety. What abolishes the person is the loop of length one, the circle of radius zero. A circle of two is a family; a circle of one is a point.

David Gerrold's *The Man Who Folded Himself* (1973) tests the boundary of the critical class. Its protagonist fathers himself with Diane, a female version of himself met in an alternate timeline. On the Everettian reading of §12.5, Diane is a counterpart rather than a stage, and the loop is heterogenetic: the

child derives from an other who is qualitatively like the father. The narrator nonetheless counts every version as himself, and his solitude follows from that count. The Heinlein Limit can therefore be reached by semantic posture as well as by topology. A subject who refuses to count his counterparts as others closes a loop that physics left open, and Korzybski's indexing of selves (Ch. 2.1) is the corresponding therapy.

Heinlein's own *Time Enough for Love* (1973) supplies a further control. Lazarus Long travels to the Kansas City of his childhood and consummates a mutual attraction with his mother, Maureen, years after his own birth. She is an adult and already pregnant, so the union produces no child: descent remains well-founded, and Lazarus's autogenetic fraction is zero. The novel states that Lazarus loved time so much that he became his own ancestor, yet no published book makes him one. It surrounds him instead with a polyamorous family that includes two female clones grown from his cells, raised as distinct persons, and it ends with that family reaching back through time to rescue him, mortally wounded, from the Western Front. Union with a progenitor and sex-inverted copies of the self, both materials of the 1959 story, here coexist with abaliety intact. Fourteen years after the story, Heinlein recombined its elements with the loop left open, and the solitude did not follow. The man who meets his origin is rescued by others; the narrator who is his own origin is left alone in the dark.

The discipline this typology inaugurates, **chronogenealogy**, asks three questions of any kinship structure in a spacetime admitting closed timelike curves: whether descent is well-founded; what each person's autogenetic fraction is; and whether every person retains at least one relation of origin terminating in an other. The third question is the criterion of personhood that the Abaliety Thesis proposes.

The four subsidiary theses announced in the Introduction now stand demonstrated. The **Heinlein Limit** is the condition $A(x) = \{x\}$, at which derivation, address, and natality vanish together. The **Apostrophic Inversion** is the perceptual form that other persons assume for a subject at that limit. The **Fallacy of Autogenetic Division**, with its converse, is the deletion of the qualifier that confines self-origination to pluralities. **Chronogenealogy** is the formal discipline that measures how far any subject has approached the limit. Together they yield a definition: a person is a being who comes from another, is addressed by another, and can begin among others. Its moral corollary, drawn in Chapter 9.5, is that a subject who counts every stage of him-

self as one person turns wrongs between those stages into self-relations, so that the one he wronged disappears from view.



WHERE DID ALL YOU ZOMBIES COME FROM?

The narrator's closing question has an answer, and the answer is the definition of the person: we came from one another.

The question was posed in a year that had already begun to answer it. Hannah Arendt's *The Human Condition* (1958) opens with the launch of Sputnik in October 1957 and reads, in the public's relief at a first step toward escape from imprisonment to the earth, a deeper rebellion. Modern man, she writes, seems possessed by a revolt against human existence as it has been given—"a free gift from nowhere (secularly speaking)"—which he wishes to exchange for something he has made himself; she cites the hope of creating life in the test tube among its symptoms. Written in the summer of the year in which Arendt's book appeared, Heinlein's story is the fable of that exchange. Its heroine wanted the stars, and W.E.N.C.H.E.S. was to be her road to them. She was diverted into the one escape from the human condition more radical than leaving the earth: leaving natality, by becoming self-made in the literal sense. The story completes the exchange Arendt described and shows what is received in return—a life without a giver, and therefore a life that was never a gift.

Each discipline brought to bear on the story has returned the same verdict in its own idiom. Physics permits the loop and cannot explain the people outside it (Ch. 1). Heinlein's own masters—Korzybski, Twain, Cabell—supply both the narrator's premises and his cure (Ch. 2). Theology foresaw the self-caused creature and judged it impossible, divine, or satanic, and the story confirms all three verdicts at once (Ch. 3). The ancient icons of closure promise plenitude and deliver poverty (Ch. 4). The narrative form confines every address within the self until its final lines (Ch. 5). Biology forbids the body that the loop requires, while the history of mid-century medicine and adoption inscribes that body with real wounds (Ch. 6). Psychoanalysis recognizes the structure as psychosis, an inverted Schreber (Ch. 7). The metaphysics of identity confirms the narrator's identity and diagnoses its cost (Ch. 8). The philosophy of freedom finds him responsible for everything and able to forgive nothing (Ch. 9). The problem of other minds discloses his epithet as a self-description (Ch. 10). The comparative field shows that the same materials, in other hands, ground plurality (Ch. 11). The Abaliety Thesis gathers these verdicts into a single theory (Ch. 12).

The epigraphs can now be read in reverse. Aquinas held that nothing can be its own efficient cause, since it would then be prior to itself; the story shows that such priority is not impossible but uninhabitable. Milton's Satan claims to be self-begot; Heinlein makes the claim true and finds that its truth is Satan's hell, the self with no exit from itself. Ovid's Narcissus discovers that possessing what he desires has made him poor; the narrator possesses

every member of his family and is the poorest being in literature. Dante's Virgin, daughter of her son, is the creature through whom the Creator enters a lineage of others: the Gospels attach to her son a genealogy of many generations (Matthew 1; Luke 3), and her paradox opens the circle of creator and creature onto the whole human family. Jane, daughter of herself, closes the circle on one. Her genealogy has a single entry.

One relation, however, the loop could not close. Hofstadter observed of Escher's *Drawing Hands* that the two hands, each drawing the other, are sustained by a third that the picture does not show: the hand of Escher, an inviolate level outside the tangle. The narrator's world has such a level. He came from another, Robert A. Heinlein, who wrote him in a single day in July 1958; he exists only when addressed by another, the reader who opens the story. A fictional character is the purest abalietal being, wholly derived from an author and alive only in a reader's attention. The narrator's denial of others is refuted by its own reception. His apostrophe is the moment the text turns to its only other, and in reading it, the reader answers it.

To his question, then, there is an answer that he cannot give and that every reader can: from one another—from parents who were not ourselves, from languages we did not invent, from those who addressed us before we could answer, and toward those for whom we will be a beginning. To his claim that he alone exists, alone in the dark, the reader who has followed him there is the reply. The zombie who begot himself has, after all, one relation his loop could not absorb. He is read.



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KEYWORDS

PRIMARY TEXT AND AUTHOR

Robert A. Heinlein; "—All You Zombies—"; *Predestination*; American science fiction; *The Magazine of Fantasy and Science Fiction*

TERMS COINED IN THIS DISSERTATION

Abaliety Thesis; Heinlein Limit; Apostrophic Inversion; Fallacy of Autogenetic Division; Autogenetic Hypostasis; Law of the Qualifier; triad of the newcomer; autogenetic fraction; chronogenealogy; genealogical Quine atom; autonarratee; σ-topology; Wheelerian surplus; chronobiological antinomy; token-reflexive replicator; open derivation

TIME, PHYSICS, AND COSMOLOGY

time travel; causal loop; bootstrap paradox; predestination paradox; closed timelike curves; Novikov self-consistency principle; chronology protection; Gödel universe; Minkowski spacetime; block universe; eternalism; A-series and B-series; personal time; backward causation; bilking argument; one-electron universe; many-worlds interpretation; knowledge paradox

THEOLOGY AND MYTH

aseity; abaliety; causa sui; autogenesis; Trinitarian personhood; relations of origin; modalism; Richard of St. Victor; Sethian Gnosticism; Autogenes; Marian paradox; ouroboros; androgyne; autochthony; Tiresias; Narcissus

IDENTITY, LOGIC, AND MIND

personal identity; psychological continuity; four-dimensionalism; perdurantism; Parfitian reductionism; idem and ipse; non-well-founded set theory; strange loops; quines; self-reproducing automata; autopoiesis; solipsism; problem of other minds; philosophical zombies

FREEDOM, RECOGNITION, AND EXISTENCE

fatalism; predestination; eternal recurrence; amor fati; compatibilism; Frankfurt cases; recognition; Aufforderung; natality; plurality; thrownness; despair; the gift

PSYCHOANALYSIS

family romance; rescue fantasy; primal scene; Nachträglichkeit; the uncanny; the double; repetition compulsion; death drive; mirror stage; foreclosure; Name-of-the-Father; sinthome; Schreber; Anti-Oedipus

BODY, SEX, AND BIOPOLITICS

intersex studies; transgender studies; sex reassignment; gender performativity; passing; one-sex model; sexual difference; biopower; Baby Scoop Era; coefficient of inbreeding; de novo mutation; Muller's ratchet

NARRATIVE AND FORM

fabula and syuzhet; chronotope; confession genre; Augustinian narration; narrative levels; metalepsis; interpellation; retrograde form

SOCIETY, POLITICS, AND INFORMATION

possessive individualism; self-ownership; the self-made man; the organization man; incest prohibition; exogamy; Leviathan; political theology; model collapse; recursive training; filter bubbles

FIELDS

speculative fiction; philosophical fiction; science fiction studies; comparative literature; time-travel narrative; philosophy and literature



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